Spectral chirality of lens spaces: level-wise helicity balance, eta congruences, and silver zeros for transverse-traceless tensor curl

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Abstract

Let A_J be the rank-J projected transverse-traceless tensor-curl operator on a round lens space $L(p,q)$, in the normalization of [1], with helicity branches $V_{n+2J} \otimes V_n$ and $V_n \otimes V_{n+2J}$ and eta invariant $\eta_J(L(p,q))$. We study the finer question of whether the two branches have equal multiplicity at every level, and the arithmetic of η_J that the answer leaves behind. We write the level-wise helicity imbalance as an explicit sum over group elements with a closed-form trigonometric kernel, prove that balance at every level is equivalent to a finite identity in the group ring $\mathbb{Z}[\mathbb{Z}/p]$ together with the absence of noncentral Clifford-translation elements, and classify all balanced triples (p,q,J) with $p \geq 3$: they are the mirror-symmetric spaces $q^2 \equiv -1 \pmod p$, the "silver" family $p \mid 4J^2-2$ with $q \equiv \pm(2J \pm 1)$, and the blind ranks $p \mid 2J$ under the Clifford condition. At rank one this recovers, by composition of classical results, the statement that the coexact curl spectrum of $L(p,q)$ is symmetric level by level exactly when $L(p,q)$ is amphicheiral. We then prove a divisor decomposition of $p\,\eta_J$ into primitive parts, the congruence $\eta_J - J(2J^2-1)\eta_1 \in 2\mathbb{Z}$ at every modulus, the sharper prime-modulus congruence $p\,\eta_J \equiv -J(2J^2-1)(q+q^{-1})/3 \pmod p$ with $p\,\eta_J \in 2\mathbb{Z}$ for primes $p \geq 5$, and the resulting exact denominators. These settle the zero set of η_J at prime moduli $p \geq 5$ away from the silver ranks $2J^2 \equiv 1 \pmod p$: for $p \nmid J$ the zeros are exactly the mirror spaces. At silver ranks η_J is an even integer, and we prove that $\eta_J(L(p,J)) = 0$ for every divisor $p > 1$ of $2J^2-1$, extending a family of [1] from $p = 2J^2-1$ to all its divisors; the mechanism is the conjugacy invariance of Rademacher's symbol Ψ on a trace -6 matrix commuting with an integral matrix of determinant -1. Finally we record what a vanishing Rademacher symbol does and does not detect: inert and reciprocal hyperbolic classes force $\Psi = 0$, neither is necessary, and the matrices behind the silver zero lie in the inert class R^2L^2. The zero set at the level $p = 2J^2-1$ itself (silver S-units, a fifth family, a Pell ladder and a period law) is the subject of a companion paper [2], whose main statements we summarize. The complete zero set at silver ranks remains open.

1 Introduction

The paper [1] evaluates the eta invariant $\eta_J(L(p,q))$ of the rank-J projected transverse-traceless (TT) tensor-curl operator on every round lens space, uniformly in the rank and in prime and composite levels, and exhibits four infinite families of zeros at $p = 2J^2-1$. It leaves open the classification of all zero orbits "beyond the four constructions above and the orientation-forced condition $q^2 \equiv -1 \pmod p$". The eta invariant is a regularized sum of the level-wise imbalance $D_J(n) = N_J^+(n) - N_J^-(n)$ between the two helicity branches; a vanishing eta invariant may or may

not come from a vanishing imbalance at every level. The present paper is a companion to [1]. It uses the same operator, the same normalization and the same lens-space conventions, and it addresses three questions in turn.

(i) *Level-wise balance.* For which (p, q, J) is $D_J(n) = 0$ for every level $n \geq 0$? Sections 3 and 4 answer this completely for $p \geq 3$ (Theorems 3.1, 3.3 and 4.1). At rank one the answer is classical by composition of three published results [8, 19, 24, 23, 11]; for $J \geq 2$ we found neither the criterion nor the families in the literature we searched.

(ii) *Congruences.* Sections 5 and 6 prove a decomposition of $p\,\eta_J$ over the divisors of p, the congruence $\eta_J - J(2J^2-1)\eta_1 \in 2\mathbb{Z}$ at every modulus (Theorem 6.7), and at primes $p \geq 5$ the finer statement $p\,\eta_J \in 2\mathbb{Z}$, $p\,\eta_J \equiv -J(2J^2-1)(q+q^{-1})/3 \pmod p$ (Theorem 6.3). These determine the exact denominator of η_J and settle the zero problem of [1] at prime moduli $p \geq 5$ away from the silver ranks $2J^2 \equiv 1 \pmod p$.

(iii) *Silver zeros.* At a silver rank η_J is an even integer and the zero set is richer. Section 7 proves $\eta_J(L(p, J)) = 0$ for every divisor $p > 1$ of $2J^2 - 1$ (Theorem 7.1), the first family of [1] extended from $p = 2J^2 - 1$ to all its divisors. Section 8 explains the mechanism in the language of Rademacher's symbol on hyperbolic conjugacy classes and records, with citations and without novelty claims, what a vanishing symbol detects and what it forgets. Section 9 summarizes what the companion paper [2] proves about the zero set at $p = 2J^2 - 1$: the silver S-unit coordinate, a fifth zero family, the Pell ladder, the period law for every orbit of norm ± 1 or ± 2, and the unbounded ambiguity of a vanishing η_J.

Relation to the companion paper [3]: that paper transfers twisted spin-Dirac eta vectors to all-rank TT eta vectors on every free round spherical space form, and organizes the one-sided quotients along affine-ADE diagrams. It contains no level-wise statements, no congruences and no zero classification, and it is not used here; conversely nothing below concerns non-cyclic quotients. The boundary realization of the operator in [4] is likewise not used.

Helicity-resolved tensor mode counts on lens spaces are also of independent interest in cosmic topology, where the effect of topology on spin-two eigenmodes was recently described as unstudied [15]; no physical claim is made in this paper.

Exact computations are recorded in Appendix A. They test the identities and conventions; they do not replace the proofs of universal statements.

2 Conventions

We keep the conventions of [1, Sec. 2] verbatim. Write $S^3 = \{(z_1, z_2) \in \mathbb{C}^2 : |z_1|^2 + |z_2|^2 = 1\}$. For $p \geq 2$ and $q \in (\mathbb{Z}/p\mathbb{Z})^\times$ the generator of $\mathbb{Z}_p$ acts by

$$(z_1, z_2) \longmapsto (\mathrm{e}^{2\pi i/p} z_1, \mathrm{e}^{2\pi i q/p} z_2),$$

and $L(p, q)$ is the quotient with its complex quotient orientation. We write q^* for the inverse of q modulo p, and $\tilde{q}$ for the odd lift of q ($\tilde{q} = q$ if q is odd, $\tilde{q} = q + p$ if q is even and p odd; if p is even every unit is odd). Put

$$A = \frac{1+\tilde{q}}{2}, \qquad B = \frac{1-\tilde{q}}{2}, \qquad A + B = 1. \tag{1}$$

Let V_a be the irreducible $SU(2)$ representation of highest weight a and dimension $a+1$, with weights $W_a = \{-a, -a+2, \ldots, a\}$. The lifted generator acts on the weight vector (w_L, w_R) of $V_a \otimes V_b$

by $\exp(2\pi i(Aw_L + Bw_R)/p)$; by [1, Lem. 5.2] (central descent) the invariant counts below do not depend on the lift. For $a \equiv b \pmod 2$ this is the cyclic invariant count $I_{a+1,b+1}(p,q)$ of [1, Sec. 2],

$$N(a,b) = \#\{(w_L, w_R) \in W_a \times W_b : \ Aw_L + Bw_R \equiv 0 \pmod p\} = I_{a+1,b+1}(p,q). \tag{2}$$

The rank-J projected TT tensor-curl operator A_J of [1, Thm. 3.1] has, on the unit sphere, the signed spectrum

$$V_{n+2J} \otimes V_n : \ +(n+J+1), \qquad V_n \otimes V_{n+2J} : \ -(n+J+1), \qquad n \geq 0,$$

and on $L(p,q)$ the eigenspaces are the invariant vectors. For a *level* $n \geq 0$ put

$$N_J^+(n) = N(n{+}2J, n), \qquad N_J^-(n) = N(n, n{+}2J), \qquad D_J(n) = D_J(n;p,q) = N_J^+(n) - N_J^-(n), \tag{3}$$

and write $\nu = n + J + 1$ for the eigenvalue magnitude. The eta invariant is

$$\eta_J(L(p,q)) = \eta_{A_J}(0), \qquad \eta_{A_J}(s) = \sum_{n \geq 0} \frac{D_J(n)}{(n+J+1)^s}, \tag{4}$$

continued to $s = 0$ as in [1, Prop. 4.1]; the continuation is regular at $s = 0$ and its value equals the Abel value of the corresponding root-of-unity exponential-polynomial series [1, Lem. 5.4]. We use the lens symmetries

$$\eta_J(L(p,q^*)) = \eta_J(L(p,q)), \qquad \eta_J(L(p,-q)) = -\eta_J(L(p,q)), \tag{5}$$

the rank-one value $\eta_1(L(p,q)) = -4s(q,p)$ [1, Cor. 5.8], where

$$((x)) = \begin{cases} x - \lfloor x \rfloor - \frac{1}{2}, & x \notin \mathbb{Z}, \\ 0, & x \in \mathbb{Z}, \end{cases} \qquad s(q,p) = \sum_{a=0}^{p-1} \left(\!\left(\frac{a}{p}\right)\!\right) \left(\!\left(\frac{qa}{p}\right)\!\right),$$

and the all-rank master formula of [1, Thm. 5.1],

$$\begin{aligned} \eta_J(L(p,q)) = \ & -4J\, s(q,p) + \frac{2}{p} \sum_{a=1}^{J-1} (J-a)\big[\rho_p(aq) + \rho_p(aq^*)\big] \\ & - 2 \sum_{\substack{a,b \geq 1 \\ a+b \leq J-1}} (J-a-b)\big[\delta_p(aq-b) - \delta_p(aq+b)\big] - L_J(p,q), \end{aligned} \tag{6}$$

with $\rho_p(k) = 0$ if $p \mid k$ and $p - 2\langle k \rangle_p$ otherwise, $\delta_p(k) = \mathbf{1}_{p|k}$, and the low-mode term $L_J(p,q) = \sum_{m=1}^{J-1} [I_{J-m,J+m}(p,q) - I_{J+m,J-m}(p,q)]$. Equation (6) holds for every $p \geq 2$, every unit q and every $J \geq 1$. Its last two terms combine into a single signed sum over lattice points, the box form of [2, Lem. 8.1]; that is a rewriting of these two terms, not a new invariant, and nothing in this paper uses it.

Definition 2.1. Let $p \geq 1$, q a unit modulo p and $J \geq 1$.

(a) $L(p,q)$ is *rank-J balanced* (balanced at every level) if $D_J(n;p,q) = 0$ for all $n \geq 0$.

(b) $L(p,q)$ is *mirror* (amphicheiral) if $q^2 \equiv -1 \pmod p$; otherwise it is *chiral*. The condition is the classical criterion for an orientation-reversing self-homeomorphism, indeed an orientation-reversing isometry [23, 11]: if $q^2 \equiv -1$ then $T(z_1, z_2) = (\bar{z}_2, z_1)$ normalizes the deck group and reverses orientation.

(c) The rank J is a *silver rank* for p if $2J^2 \equiv 1 \pmod p$ (equivalently $p \mid 2J^2-1$, so p is odd and coprime to 6).

(d) The rank J is a *blind rank* for p if $p \mid 2J$.

(e) The *Clifford condition* is

$$\gcd(q-1,p) \le 2 \quad \text{and} \quad \gcd(q+1,p) \le 2. \tag{C}$$

For odd p it says $\gcd(q^2-1,p)=1$; for even p it says that both gcds equal 2. It has nonempty scope only when $\gcd(p,12) \in \{1,2\}$ (Proposition 3.8).

The mirror condition and the two lens symmetries (5) give the two "forced" facts that anchor everything below: a mirror space has $D_J(n) = 0$ at every level and every rank, because the orientation-reversing isometry exchanges the two branches; and $\eta_J(L(p,q)) = 0$ whenever $q^* \equiv -q$, by (5).

Notation for group elements. For $0 \le k < p$ put $\theta_k = 2\pi k/p$. Let

$$D^+ = \{k:\ p \mid k(q-1)\}, \qquad D^- = \{k:\ p \mid k(q+1)\}, \qquad Z = \{k:\ 2k \equiv 0 \pmod p\}, \tag{7}$$

so that $Z = \{0\}$ or $\{0, p/2\}$, and put $D^{\pm *} = D^\pm \setminus Z$ and $\mathcal{N} = \mathbb{Z}/p \setminus (D^+ \cup D^-)$. The element g^k with $k \in D^{+*}$ (resp. D^{-*}) acts on S^3 as a pure left (resp. right) Clifford translation; we call these the *Clifford-translation elements*. Condition (C) says exactly that $D^{+*} = D^{-*} = \emptyset$, since D^+ is the subgroup of order $\gcd(q-1,p)$ and D^- the subgroup of order $\gcd(q+1,p)$. Finally $\chi_a(t) = \sin((a+1)t)/\sin t$ is the $SU(2)$ character (with its removable limits $\chi_a(0) = a+1$, $\chi_a(\pi) = (-1)^a(a+1)$),

$$c_J(\nu,t) = \frac{2\big[\nu\cos(\nu t)\sin(Jt) - J\sin(\nu t)\cos(Jt)\big]}{\sin t}, \qquad \mathcal{K}_J(t) = \frac{\sin(Jqt)}{\cos(qt)-\cos t} + \frac{\sin(Jq^*t)}{\cos(q^*t)-\cos t}, \tag{8}$$

and for $a \in \mathbb{Z}/p$ we write $[a] = \delta_a - \delta_{-a} \in \mathbb{Z}[\mathbb{Z}/p]$, so that $[a] = 0$ iff $2a \equiv 0$.

3 Level-wise helicity balance

3.1 The imbalance as a sum over group elements

The per-element signed character of [1, Sec. 5.1] is, for a nonidentity element g^k and $\nu = n+J+1$,

$$E_k(n) = \chi_{n+2J}(x_k)\chi_n(y_k) - \chi_n(x_k)\chi_{n+2J}(y_k), \qquad x_k = \frac{2\pi kA}{p}, \quad y_k = \frac{2\pi kB}{p}, \tag{9}$$

with $u_k := x_k - y_k = 2\pi k\tilde{q}/p \equiv q\theta_k$ and $v_k := x_k + y_k = \theta_k$ modulo 2π. Character averaging gives $pD_J(n) = \sum_{k=0}^{p-1} E_k(n)$. The following theorem reorganizes this sum. Its content is the closed-form kernel $\mathcal{K}_J$ and the explicit separation of the Clifford-translation elements; the per-element character itself is from [1], and at $J=1$ the generating-function form of the same sum is in [8, Thm. 5.5, Cor. 5.8, Ex. 5.9].

Theorem 3.1 (Kernel form of the level-wise imbalance)**.** *For $p \ge 1$, every unit q, $J \ge 1$ and $n \ge 0$, with $\nu = n+J+1$,*

$$p\,D_J(n;p,q) = \sum_{k\in D^{+*}} c_J(\nu,\theta_k) - \sum_{k\in D^{-*}} c_J(\nu,\theta_k) + 2\sum_{k\in\mathcal{N}} \sin(\nu\theta_k)\,\mathcal{K}_J(\theta_k). \tag{10}$$

The third sum is the generic part of $\sum_k E_k$ after the reindexing $k \mapsto q^*k$ of the second half of each character difference; its k-th summand is not in general the single-element contribution $E_k(n)$.

Proof. Step 1. For $k \in Z$, $E_k = 0$: at $k = 0$ the two dimensions agree, and at $k = p/2$ (even p) both x_k and y_k lie in $\pi\mathbb{Z}$, so the character values are $(-1)^{\text{weight}}\times$ dimension and the two products agree ([1, Lem. 5.3]).

Step 2 (generic elements). Since $\sin x \sin y = \frac{1}{2}(\cos u - \cos v)$ with $u = x - y$, $v = x + y$, the denominator of E_k vanishes iff $u_k \equiv \pm v_k \pmod{2\pi}$ iff $k \in D^+ \cup D^-$. For $k \in \mathcal{N}$, expanding the numerator by product-to-sum with $a = \nu + J$, $b = \nu - J$ gives

$$\sin(ax)\sin(by) - \sin(bx)\sin(ay) = \sin(\nu v)\sin(Ju) - \sin(\nu u)\sin(Jv),$$

hence

$$E_k(n) = \frac{2[\sin(\nu v_k)\sin(Ju_k) - \sin(\nu u_k)\sin(Jv_k)]}{\cos u_k - \cos v_k} = \frac{2\sin(\nu\theta_k)\sin(Jq\theta_k)}{\cos(q\theta_k) - \cos\theta_k} - \frac{2\sin(\nu q\theta_k)\sin(J\theta_k)}{\cos(q\theta_k) - \cos\theta_k}. \tag{11}$$

Step 3 (reindexing). Multiplication by the unit q preserves D^+, D^- and $\mathcal{N}$. Substituting $k = q^*k'$ in the second term of (11) and summing over $k \in \mathcal{N}$ gives $\sum_{k' \in \mathcal{N}} 2\sin(\nu\theta_{k'})\sin(Jq^*\theta_{k'})/(\cos(q^*\theta_{k'}) - \cos\theta_{k'})$. Adding the first term yields $2\sum_{\mathcal{N}} \sin(\nu\theta_k)\mathcal{K}_J(\theta_k)$.

Step 4 (Clifford translations). For $k \in D^{+*}$ one has $y_k \in \pi\mathbb{Z}$, say $y_k = \pi j$, and $\sin\theta_k \neq 0$. With $\sigma = (-1)^j$, $\chi_a(y_k) = \sigma^a(a+1)$ and $\chi_a(x_k) = \chi_a(\theta_k - \pi j) = \sigma^a\chi_a(\theta_k)$, so

$$\begin{aligned} E_k(n) &= (n+1)\chi_{n+2J}(\theta_k) - (n+2J+1)\chi_n(\theta_k) \\ &= \frac{(\nu - J)\sin((\nu+J)\theta_k) - (\nu+J)\sin((\nu-J)\theta_k)}{\sin\theta_k} = c_J(\nu, \theta_k). \end{aligned}$$

For $k \in D^{-*}$, $x_k \in \pi\mathbb{Z}$ and symmetrically $E_k(n) = -c_J(\nu, \theta_k)$. Since $D^+ \cap D^- \subseteq Z$, the three sums in (10) run over disjoint sets. □

Corollary 3.2 (Affine behaviour on residue classes; finite window)**.** *Write $p\,D_J(n) = \nu F(\nu) + G(\nu)$ with*

$$F(\nu) = \sum_k \varphi(k)\cos(\nu\theta_k), \qquad G(\nu) = \sum_k \psi(k)\sin(\nu\theta_k), \tag{12}$$

where $\varphi = \dfrac{2\sin(J\theta)}{\sin\theta}(\mathbf{1}_{D^{+*}} - \mathbf{1}_{D^{-*}})$ is even and $\psi = -\dfrac{2J\cos(J\theta)}{\sin\theta}(\mathbf{1}_{D^{+*}} - \mathbf{1}_{D^{-*}}) + 2\mathcal{K}_J\mathbf{1}_{\mathcal{N}}$ is odd in k. Both F and G are p-periodic in ν. Consequently for each $0 \le r < p$ there are integers u_r, v_r with

$$D_J(r + pk; p, q) = u_r k + v_r \qquad (k \ge 0), \tag{13}$$

and $L(p, q)$ is rank-J balanced iff $D_J(n) = 0$ for $0 \le n < 2p$.

Proof. Read off from (10) and (8). Two values of k determine an affine function. □

The affine statement on classes modulo $2p$, which already gives the window $n < 4p$, is [1, Prop. 4.1]; the sharper modulus p can also be obtained directly by an elementary character argument (the increment $c_{h+p} - c_h$ of the residue-class weight counts depends only on h modulo 2).

3.2 The criterion

Theorem 3.3 (Complete criterion for level-wise balance)**.** *Let $p \geq 3$, q a unit modulo p and $J \geq 1$. Then $L(p, q)$ is rank-J balanced if and only if both*

(i) the Clifford condition (C) holds, i.e. $\gcd(q-1, p) \leq 2$ *and* $\gcd(q+1, p) \leq 2$*; and*

(ii) in $\mathbb{Z}[\mathbb{Z}/p]$, with $X = Jq$ and $Y = Jq^*$,

$$[X + q^*] + [X - q^*] + [Y + q] + [Y - q] = [X + 1] + [X - 1] + [Y + 1] + [Y - 1]. \tag{14}$$

For $p \in \{1, 2\}$ every $L(p, q)$ is balanced at every rank.

Proof. Necessity. Suppose $D_J(n) = 0$ for all $n \geq 0$. Each residue class of ν modulo p contains infinitely many $\nu \geq J + 1$, and $\nu F(\nu) + G(\nu)$ is affine on the class, so $F = G = 0$ on $\mathbb{Z}/p$. Since φ is even and ψ is odd, F is the full Fourier transform of φ and G that of ψ (up to the factor i), so Fourier inversion gives $\varphi = \psi = 0$.

- $\varphi = 0$ gives $\sin(J\theta_k) = 0$ on $D^{+*} \cup D^{-*}$ (the two sets are disjoint, as k in both forces $p \mid 2k$), hence $|\cos(J\theta_k)| = 1$ there.
- Then $\psi = \mp 2J \cos(J\theta_k)/\sin\theta_k \neq 0$ on $D^{\pm *}$, contradicting $\psi = 0$ unless $D^{+*} = D^{-*} = \emptyset$. Since $D^{\pm}$ is the subgroup of order $\gcd(q \mp 1, p)$, this is (i).
- Finally $\psi = 0$ on $\mathcal{N}$ says $\mathcal{K}_J(\theta_k) = 0$ for $k \in \mathcal{N}$, which is (ii) by the lemma below.

Sufficiency. Under (i), $D^{\pm *} = \emptyset$ and $pD_J(n) = 2\sum_{\mathcal{N}} \sin(\nu\theta_k)\mathcal{K}_J(\theta_k)$, which vanishes identically when $\mathcal{K}_J = 0$ on $\mathcal{N}$, i.e. under (ii).

Lemma. Under (i), $\mathcal{K}_J = 0$ on $\mathcal{N}$ iff (14). Indeed, on $\mathcal{N}$ both denominators of $\mathcal{K}_J$ are nonzero, so $\mathcal{K}_J(\theta) = 0$ iff

$$S(\theta) := \sin(X\theta)\big(\cos(q^*\theta) - \cos\theta\big) + \sin(Y\theta)\big(\cos(q\theta) - \cos\theta\big) = 0.$$

Product-to-sum gives $2S(\theta) = \sum_{a \in P} \sin(a\theta) - \sum_{b \in M} \sin(b\theta)$ with $P = (X + q^*, X - q^*, Y + q, Y - q)$ and $M = (X + 1, X - 1, Y + 1, Y - 1)$. At $k \in Z$ (where $\theta \in \{0, \pi\}$) S vanishes automatically, and under (i) $\mathcal{N} \cup Z = \mathbb{Z}/p$; so $\mathcal{K}_J = 0$ on $\mathcal{N}$ iff $S(\theta_k) = 0$ for all $k \in \mathbb{Z}/p$. Now S is the sine transform of the odd measure $\mu = \sum_P [a] - \sum_M [b]$: $\sum_r \mu(r) e^{ir\theta} = 4iS(\theta)$. By Fourier inversion $S \equiv 0$ iff $\mu = 0$, which is (14).

For $p = 1$ the two branches have equal dimension; for $p = 2$ both branch weights have the parity of n and the invariant counts agree. □

3.3 Consequences and the three sufficient families

Corollary 3.4. *Let $p \geq 3$.*

(a) (Descent.) *If $L(p, q)$ is rank-J balanced, so is $L(d, q \bmod d)$ for every $d \mid p$.*

(b) (Rank periodicity and reflection, level-wise.) *The rank-J verdict depends on J only modulo p, and J and $-J$ give the same verdict.*

(c) (Mirror.) *If $q^2 \equiv -1 \pmod{p}$, then $L(p, q)$ is rank-J balanced for every J.*

(d) (Silver family.) *If $p \mid 4J^2 - 2$ and $q \equiv \pm(2J - 1)$ or $q \equiv \pm(2J + 1) \pmod{p}$, then $L(p, q)$ is rank-J balanced and chiral.*

(e) (Blind ranks.) *If $p \mid 2J$ and (C) holds, then $L(p,q)$ is rank-J balanced.*

(f) If $L(p,q)$ is rank-J balanced, then $\eta_J(L(p,q)) = 0$.

Proof. (a) Condition (i) passes to divisors, and $S = 0$ on $\mathbb{Z}/p$ restricts to $S = 0$ on the subgroup $(p/d)\mathbb{Z}/p$, which is the function S of $L(d, q \bmod d)$. (b) (14) involves J only through X, Y modulo p, and $J \mapsto -J$ sends $\mu \mapsto -\mu$. (c) If $d \mid \gcd(q \mp 1, p)$ then $q^2 + 1 \equiv 2 \pmod{d}$, so $d \leq 2$, and (i) holds; moreover $q^* \equiv -q$, so $Y \equiv -X$ and the four P-terms cancel in pairs, as do the M-terms. (Alternatively, the orientation-reversing isometry exchanges the branches.) (d) The verdict is invariant under $q \mapsto -q$ (which negates μ) and $q \mapsto q^*$ (which exchanges X and Y), so take $q = 2J - 1$; then $q^* = 2J + 1$ because $(2J-1)(2J+1) = 2(2J^2-1)+1$. Put $e = 2J^2 - 1$ (odd) and $c = 0$ if p is odd, $c = p/2$ if p is even. If p is even then $p = 2d$ with $d \mid e$ odd and $e \equiv d \pmod{2d}$; in both cases $X = J(2J-1) = e - J + 1 \equiv c + 1 - J$ and $Y = J(2J+1) \equiv c + 1 + J$. Hence $P = (c + J + 2,\ c - 3J,\ c + 3J,\ c + 2 - J)$ and $M = (c + 2 - J,\ c - J,\ c + J + 2,\ c + J)$. Using $[c - a] = -[c + a]$ (as $2c \equiv 0$), the terms $[c - 3J] + [c + 3J]$ and $[c - J] + [c + J]$ vanish and the remaining terms match, proving (14). Condition (i): $\gcd(2J - 2, 2e) = \gcd(2J, 2e) = 2$ because $e = 2J \cdot J - 1$ and $e = 2(J-1)(J+1) + 1$. Chirality: $(2J-1)^2 + 1 \equiv -4(J-1) \pmod{p}$, and $p \mid 4(J-1)$ is impossible for $p \geq 3$ since the odd part of p divides e and $\gcd(e, J-1) = 1$. (e) $2X \equiv 2Y \equiv 0$ gives $[X + a] + [X - a] = 0$ for every a, so both sides of (14) vanish. (f) The eta function (4) of a balanced space vanishes identically. □

Remark 3.5. Family (d) contains, for $p = 2J^2 - 1$, the zeros $q = 2J - 1$ of [1, Thm. 7.1], which form a uniformly balanced family (balanced at every J); the other three parametrizations of that theorem are not balanced in general, although individual members can lie on the balanced or mirror loci (see Example 4.5). It also contains every proper divisor p of $2J^2 - 1$ and the even levels $p = 2d$, $d \mid 2J^2 - 1$, which lie outside that theorem ("every level p_J in these families is odd"). Some proper-divisor members are reachable from [1] by combining its zero families with its rank symmetries (for instance $\eta_5(L(7,q)) = -\eta_2(L(7,q)) = 0$ for $q \in \{2,3,4,5\}$); others are not, for instance $L(23,4)$ at $J = 9$, where $23 \mid 2 \cdot 81 - 1 = 7 \cdot 23$, no J' has $2J'^2 - 1 = 23$, and the rank symmetries only relate $J \equiv \pm 9 \pmod{23}$. Every even level $p \equiv 2 \pmod 4$ lies outside both [1, Thm. 7.1] (odd levels) and [1, Thm. 6.2] (which needs $\gcd(p, 6) = 1$).

Remark 3.6 (Mechanism of the blind ranks)**.** If $p \mid 2J$ then $\sin(Jq\theta_k) = \sin(Jq^*\theta_k) = 0$ for every k, so the kernel $\mathcal{K}_J$ vanishes at every class. This is the same vanishing of the $SU(2)$ character χ_{J-1} at every noncentral class that underlies the blind adjacent-rank families of [4], and condition (C) is the primality-free non-centrality condition used there. What is new at the level-wise scale is the *necessity* of (C) for balance, the criterion (14), and the silver family (d).

3.4 Boundedness, existence, and even levels

Proposition 3.7 (Bounded versus growing imbalance)**.** *Let $p \geq 3$. The sequence $D_J(n; p, q)$ is bounded in n iff $\gcd(q-1, p) \mid 2J$ and $\gcd(q+1, p) \mid 2J$. In that case D_J is p-periodic in n; otherwise $|D_J|$ grows linearly along at least one residue class modulo p. A Clifford-translation element (a $k \in D^{+*} \cup D^{-*}$) precludes balance at every level, but need not force growth: for $L(3,1)$ and $J = 3$ one has $D_3 = (-4, 4, 0)$ repeated with period 3, with $(N^+, N^-) = (3,7), (4,0), (9,9)$ at $n = 0, 1, 2$.*

Proof. By (12), D_J is bounded iff $F \equiv 0$ iff $\varphi = 0$ iff $\sin(J\theta_k) = 0$ on every noncentral element of D^+ and of D^-. On the subgroup D^+ of order $d = \gcd(q-1, p)$ this holds iff $d \mid 2J$, and it is automatic for $d \leq 2$; likewise for D^-. If $F = 0$ then $pD_J = G$ is p-periodic; if $F \neq 0$ some class has nonzero slope. The last assertion is Theorem 3.3: with $F = 0$ the residual G detects the Clifford elements. □

Proposition 3.8 (Existence of balanced ranks)**.** *For $p \geq 3$, some pair (q, J) with $L(p, q)$ rank-J balanced exists iff* $\gcd(p, 12) \in \{1, 2\}$*. A chiral balanced example exists iff, in addition, $p \notin \{5, 10\}$.*

Proof. If $3 \mid p$, every unit q is $\equiv \pm 1 \pmod 3$, so (C) fails; if $4 \mid p$, q is odd and 4 divides one of $q \mp 1$, so (C) fails. Conversely, for odd p coprime to 6 take $q = 2$, $J = p$; for $p = 2d$ with d odd and $3 \nmid d$ take $q = 3$, $J = d$: both satisfy (C) and $p \mid 2J$, so Corollary 3.4(e) applies. For chirality, $q = 2$ is mirror only when $p = 5$ and $q = 3$ only when $p = 10$; at $p = 5$ the admissible q are $2, 3$ and at $p = 10$ they are $3, 7$, all mirror. □

Lemma 3.9 (Even levels are a parity projection of their odd parent)**.** *Let p be odd and $\tilde{q}$ the odd lift of q. Then for every J and every level n,*

$$N_J^{\pm}(n; 2p, \tilde{q}) = \begin{cases} N_J^{\pm}(n; p, q), & n \text{ even}, \\ 0, & n \text{ odd}. \end{cases}$$

Consequently $L(2p, \tilde{q})$ is rank-J balanced iff $L(p, q)$ is. The analogous transfer of $\eta_J = 0$ fails: $\eta_3(L(17, 3)) = 0$ but $\eta_3(L(34, 3)) = -4$.

Proof. The same A, B of (1) define both actions. Since $w_L \equiv w_R \equiv n \pmod 2$ and $A + B = 1$, one has $Aw_L + Bw_R \equiv n \pmod 2$. For odd n no weight pair is invariant modulo $2p$; for even n divisibility by p is equivalent to divisibility by $2p$. If $L(p, q)$ is balanced so is its double; if the double is balanced, each residue class modulo the odd number p contains infinitely many even n, and the affine function (13) vanishing at them vanishes identically. The two eta values quoted are exact computations from (6). □

Remark 3.10. For $p = 2d$ with d odd satisfying (C), the rank-J verdict has period d in J, sharper than the period p of Corollary 3.4(b): balance at $2d$ is balance at the odd parent d, whose verdict has period d.

4 The classification of balanced triples

Theorem 4.1 (Classification of level-wise balance)**.** *Let $p \geq 3$, q a unit modulo p and $J \geq 1$. Then $L(p, q)$ has equal numbers of positive- and negative-helicity rank-J TT modes at every level if and only if one of the following holds:*

(a) (mirror) $q^2 \equiv -1 \pmod p$*;*

(b) (silver family) $p \mid 4J^2 - 2$ *and* $q \equiv \pm(2J \pm 1) \pmod p$*;*

(c) (blind rank) $p \mid 2J$ *and the Clifford condition* (C) *holds.*

Sufficiency is Corollary 3.4(c)–(e). The proof of necessity occupies the rest of this subsection. One step of it, the exhaustion of 105 finite congruence systems, is certificate-assisted: the hand reduction below covers the structure, and the complete case list was verified by integer Gröbner-basis certificates whose meaning is made explicit in Remark 4.2.

Proof of necessity. Step 0 (reductions). By Theorem 3.3, balance requires (C), which fails if $3 \mid p$ or $4 \mid p$ (Proposition 3.8). If $p \equiv 2 \pmod 4$, write $p = 2d$ with d odd. By Lemma 3.9 balance at $2d$ is equivalent to balance at d with q mod d; and each of (a), (b), (c) at $2d$ is equivalent to the same condition at d: for (a) because q is odd, so $q^2 \equiv -1 \pmod 2$ always; for (b) because $4J^2 - 2$ is even and q, $2J \pm 1$ are odd; for (c) because $2d \mid 2J$ iff $d \mid J$ iff $d \mid 2J$; and (C) at $2d$ iff at d. It therefore

suffices to treat odd $p \geq 5$ with $\gcd(q-1,p) = \gcd(q+1,p) = 1$, so that 2, q and $q \pm 1$ are units modulo p.

Step 1 (matchings). Condition (14) says that the multiset of the eight atoms

$$a_1 = X + q^*, \quad a_2 = X - q^*, \quad a_3 = Y + q, \quad a_4 = Y - q,$$
$$a_5 = -X - 1, \quad a_6 = -X + 1, \quad a_7 = -Y - 1, \quad a_8 = -Y + 1$$

($X = Jq$, $Y = Jq^*$) is invariant under negation in $\mathbb{Z}/p$. For odd p a nonzero residue is never its own negative, so negation invariance holds iff the nonzero atoms pair off as $\{a, -a\}$ and the zero atoms are even in number; that is, iff the eight labelled atoms admit a perfect matching all four of whose pair sums are $\equiv 0 \pmod{p}$. There are 105 labelled perfect matchings of eight points.

Step 2 (pair sums). Multiplying each pair sum by the unit q gives an element of $\mathbb{Z}[J, q]$:

pairs	$q\times$(pair sum)	condition after cancelling units
12, 56	$\pm 2Jq^2$	$J \equiv 0$
34, 78	$\pm 2J$	$J \equiv 0$
13	$(J+1)(q^2+1)$	
24	$(J-1)(q^2+1)$	
14	$(J-1)q^2 + (J+1)$	
23	$(J+1)q^2 + (J-1)$	
58, 67	$-J(q^2+1)$	
57	$-(J(q^2+1) + 2q)$	
68	$-(J(q^2+1) - 2q)$	
17	$(q-1)(J(q+1)-1)$	$J(q+1) \equiv 1$
18	$(q+1)(J(q-1)+1)$	$J(q-1) \equiv -1$
27	$(q+1)(J(q-1)-1)$	$J(q-1) \equiv 1$
28	$(q-1)(J(q+1)+1)$	$J(q+1) \equiv -1$
35	$(q-1)(q - J(q+1))$	$J(q+1) \equiv q$
36	$(q+1)(q - J(q-1))$	$J(q-1) \equiv q$
45	$-(q+1)(J(q-1)+q)$	$J(q-1) \equiv -q$
46	$-(q-1)(J(q+1)+q)$	$J(q+1) \equiv -q$
15, 16, 25, 26, 37, 38, 47, 48	$\pm(q-1)$, $\pm(q+1)$, $\pm q(q \pm 1)$	a unit: cannot vanish

The eight "forbidden" pairs in the last row are units under the standing hypotheses, so any matching containing one of them is contradictory. This leaves 33 matchings.

Step 3 (case analysis). Call a pair ij a *P–N* pair if $i \leq 4 < j$.

- *Case A: no P–N pair (9 matchings).* The pairs are drawn from $\{12, 13, 14, 23, 24, 34\}$ and $\{56, 57, 58, 67, 68, 78\}$. Each system reduces, by unit-coefficient linear combinations of the rows of the table, to $J \equiv 0$, to $q^2 + 1 \equiv 0$, or to a contradiction of the form $2q \equiv 0$, $4q \equiv 0$ or $q^2 \equiv 1$.

- *Case B: four P–N pairs (4 matchings).* Avoiding the forbidden pairs, the four cross pairs are one of $17, 18, 27, 28$ paired with one of $35, 36, 45, 46$, and the complementary pairs. Each system contains $J(q \pm 1) \equiv 1$ together with $J(q \pm 1) \equiv -1$ for the same sign: contradiction since 2 is a unit.

- *Case C: two P–N pairs (20 matchings).* If the two cross pairs share a block (subcases C1, C2), the system contains a contradiction of the same kind as in Case B. Otherwise (subcase

C3, 16 matchings) one cross pair comes from $\{1,2\} \times \{7,8\}$ and one from $\{3,4\} \times \{5,6\}$. Their two conditions $J(q \pm 1) \equiv c_1$, $J(q \mp 1) \equiv c_2$ with $c_1, c_2 \in \{\pm 1, \pm q\}$ are of three kinds. If both involve the same factor $J(q+1)$ or $J(q-1)$ with different right sides, they force $q \equiv \pm 1$, a contradiction (eight matchings). Otherwise subtracting them gives a linear relation $2J \equiv \pm 1 \pm q$. For four matchings, combining the two conditions then gives $2J^2 \equiv 1$, which is exactly (b) with $q \equiv \pm(2J \pm 1)$. For the remaining four the linear relation is $2J \equiv 1 - q$, $1 + q$, $-1 - q$ or $-1 + q$, which does not by itself make J silver; there the two remaining same-block pairs are $(J \mp 1)K$ and JK with $K = q^2 + 1$, so their difference forces $K \equiv 0$, that is (a), with the linear relation as an additional necessary condition on that matching (Remark 4.2).

Every step divides only by 2, q and $q \pm 1$, which are units modulo every odd p satisfying (C), prime or composite. Hence whenever some matching has all four pair sums $\equiv 0 \pmod p$, one of $J \equiv 0$ (that is, $p \mid J$, so $p \mid 2J$, which is (c) since (C) holds), $q^2 \equiv -1$ (that is, (a)), or $2J^2 \equiv 1$ with $q \equiv \pm(2J \pm 1)$ (that is, (b), because p odd and $p \mid 2J^2 - 1$ give $p \mid 4J^2 - 2$) holds. □

Remark 4.2 (The certificate)**.** The hand analysis of Step 3 is a case list, and it was checked mechanically as follows. For each of the 105 labelled matchings, let $I \subset \mathbb{Z}[J, q]$ be the ideal generated by its four polynomials $q \times$ (pair sum), and let I^{sat} be its saturation by the unit $2q(q^2-1)$, computed as the elimination ideal of $I + (t \cdot 2q(q^2-1) - 1)$ in $\mathbb{Z}[J, q, t]$ by a Gröbner basis with integer coefficient domain. A matching is *certified* contradictory if $I^{\mathrm{sat}} = (1)$, and certified of class (a), (b) or (c) if the generators $q^2 + 1$, resp. $\{2J^2 - 1,\ q - s\}$ (s the named root $\pm(2J \pm 1)$), resp. J reduce to 0 modulo I^{sat} over the integers. Membership of f in I^{sat} means that $(2q(q^2-1))^N f \in I$ for some N, an identity in $\mathbb{Z}[J, q]$; reducing modulo any odd p satisfying (C) then gives $f \equiv 0 \pmod p$ whenever the four pair sums vanish. The certificate is therefore a set of *implications* (vanishing pair sums force the class generators to vanish), tested matching by matching; it is not a list of 105 exported ideal bases, and one representative basis per class (the first of its class) was retained. The certified classes are: 93 contradictions (representative basis $\{1\}$); two matchings of class (c), $\{12, 34, 56, 78\}$ and $\{12, 34, 58, 67\}$ (representative $\{J\}$); one matching, $\{13, 24, 56, 78\}$, of both classes (a) and (c) (representative $\{1 + q^2, J\}$: the ideal forces both conditions, not their union); five of class (a), $\{13, 24, 58, 67\}$, $\{17, 24, 36, 58\}$, $\{18, 24, 35, 67\}$, $\{13, 27, 46, 58\}$, $\{13, 28, 45, 67\}$ (representative $\{1 + q^2\}$, the basis of the first of them); and four of class (b), one for each root, with the bases

matching	root	certified Gröbner basis
$\{17, 23, 45, 68\}$	$q \equiv 2J - 1$	$\{q^2 + 2q - 1,\ 2J - q - 1,\ J(q+1) - 1\}$
$\{18, 23, 46, 57\}$	$q \equiv -(2J - 1)$	$\{q^2 - 2q - 1,\ 2J + q - 1,\ J(q+1) + q\}$
$\{14, 27, 35, 68\}$	$q \equiv 2J + 1$	$\{q^2 - 2q - 1,\ 2J - q + 1,\ J(q+1) - q\}$
$\{14, 28, 36, 57\}$	$q \equiv -(2J + 1)$	$\{q^2 + 2q - 1,\ 2J + q + 1,\ J(q+1) + 1\}$

Each (b) basis says exactly "q is the named root and $2J^2 \equiv 1$". The representative basis $\{1 + q^2\}$ is the saturated ideal of $\{13, 24, 58, 67\}$ only. For the four class-(a) matchings with two cross pairs the ideal also contains a linear relation between q and J:

matching	relations forced, besides $q^2 + 1 \equiv 0$
$\{17, 24, 36, 58\}$	$2J + q - 1 \equiv 0$
$\{18, 24, 35, 67\}$	$2J - q - 1 \equiv 0$
$\{13, 27, 46, 58\}$	$2J + q + 1 \equiv 0$
$\{13, 28, 45, 67\}$	$2J - q + 1 \equiv 0$

(for the first, the pairs 17 and 36 give $J(q+1) \equiv 1$ and $J(q-1) \equiv q$, whence $2J \equiv 1-q$; the pairs 24 and 58 give $(J-1)(q^2+1) \equiv J(q^2+1) \equiv 0$, whence $q^2+1 \equiv 0$). That the one-generator basis is not the full ideal of these four is witnessed at $p=5$, $q=2$, $J=5$: there $q^2+1 \equiv 0$ and $2q(q^2-1)$ is a unit, yet the pair sums of $\{17,24,36,58\}$ are $(2,0,3,0)$ modulo 5. This concerns only the description of the certificate; the implication to class (a), which is all that the classification uses, holds for all five. The semantic check of Appendix A confirms on concrete moduli that the criterion is equivalent to "(C) and some matching with vanishing pair sums", that every such matching is one of the twelve above, and that the criterion is equivalent to (a) or (b) or (c).

Elementary hand check. Exhaust the 33 matchings without a forbidden pair by hand, dividing only by the units 2, q, $q \pm 1$ and never by J or by q^2+1: the 9 matchings without a cross pair (the positive pairings $(12,34)$, $(13,24)$, $(14,23)$ against the negative pairings $(56,78)$, $(57,68)$, $(58,67)$; the first forces $J \equiv 0$ and is compatible with $(56,78)$ and $(58,67)$ but not with $(57,68)$; the second forces $q^2+1 \equiv 0$ by subtraction and is compatible with $(56,78)$ only together with $J \equiv 0$, with $(58,67)$ without restriction, and not with $(57,68)$; the third forces $2(q^2-1) \equiv 0$, impossible), giving 4 possible matchings and 5 contradictions; the 4 matchings with four cross pairs and the 4 with two cross pairs in the same pair of blocks, all contradictions (equal factors $J(q \pm 1)$ with opposite right sides); and the 16 mixed-block matchings, of which 8 force $q \equiv \pm 1$, 4 are the mirror matchings of the table above, and 4 are the silver matchings (cross pairs $(17,45)$, $(18,46)$, $(27,35)$, $(28,36)$, giving $q \equiv 2J-1$, $1-2J$, $2J+1$, $-2J-1$ with $2J^2 \equiv 1$). In all, the 12 possible matchings and 21 contradictions of the certified list, which with the 72 matchings containing a forbidden pair exhaust the 105. This gives the elementary proof of the case list; the certificate remains as mechanical corroboration.

Corollary 4.3. *Let $p \geq 3$.*

(a) A chiral lens space is rank-J balanced only through the blind ranks (c) or the silver family (b). For odd p, (b) allows at most four residue classes of J modulo p (namely $2J \equiv \pm q \pm 1$); for $p \equiv 2 \pmod 4$ the same holds modulo $p/2$.

(b) If p is odd and $L(p,q)$ is chiral and rank-J balanced, then $p \mid J$ or $\gcd(J,p) = 1$.

Proof. (a) is immediate from the theorem. (b) In case (c), $p \mid 2J$ and p odd give $p \mid J$; in case (b), $p \mid 2J^2-1$ and $\gcd(J, 2J^2-1) = 1$ give $\gcd(J,p) = 1$. □

Corollary 4.4 (Rank one)**.** *For every $p \geq 1$ and every unit q, the rank-one (coexact curl) spectrum of $L(p,q)$ is balanced at every level if and only if $q^2 \equiv -1 \pmod p$, i.e. iff $L(p,q)$ is amphicheiral.*

Proof. For $p \geq 3$ this is Theorem 4.1 at $J=1$: (b) would need $p \mid 2$ and (c) would need $p \mid 2$. For $p \leq 2$ both sides hold. A direct proof valid for all p: if $q^2 \equiv -1$ the orientation-reversing isometry T of Definition 2.1 exchanges the two branches. Conversely balance forces $\eta_1(L(p,q)) = -4s(q,p) = 0$, and $s(q,p) = 0$ iff $q^2 \equiv -1 \pmod p$ [24, p. 36]. For completeness: if $s(q,p) = 0$ then Dedekind reciprocity multiplied by $12pq$ gives $12pq\, s(p,q) = p^2+q^2+1-3pq$, whose left side is a multiple of p because $6q\, s(p,q) \in \mathbb{Z}$, so $q^2+1 \equiv 0 \pmod p$; if $q^* \equiv -q$ then $s(q,p) = s(q^*,p) = s(-q,p) = -s(q,p)$. □

This rank-one statement is classical by composition: the eta invariant of coexact curl is the signature eta (Millson, as recalled in [8]), which on $L(p,q)$ is $-4s(q,p)$ [5, 19, 18]; the Dedekind sum vanishes iff $q^2 \equiv -1$ [24, 22, 17]; and $q^2 \equiv -1$ is the amphicheirality criterion [23, 11]. The composed sentence itself we did not find in print; its nearest published relative is [11, Rem. 4.13] for the Dirac operator, proved there for prime p and conjectured for odd p. The rank-$J \geq 2$ statements of Theorems 3.3 and 4.1, and the families (b) and (c), were not found in the literature we searched;

the helicity-*summed* spin-s multiplicities on $L(p,q)$ as sums over $\mathbb{Z}_p$ of products of $SU(2)$ characters are standard [14, 9, 20]. This is a bounded statement about a search, not a claim of absence.

Example 4.5 (Zero eta without balance)**.** The four zero families of [1, Thm. 7.1] at $p = 2J^2 - 1$ are $q = J$, $J+1$, $2J-1$ and $J(J-1)$ and their orbit images. The parametrization $q = 2J-1$ (with its inverse $2J+1$ and their negatives) is uniformly balanced, at every J; the other three are not balanced in general, but their members can overlap the balanced or mirror loci, and the named families are not disjoint. For $J = 3$, $p = 17$: $\eta_3(L(17,q)) = 0$ for $q \in \{3, 6, 11, 14\}$ (the orbit of $q = J = 3$), but $L(17,3)$ has $(N_3^+, N_3^-) = (1,3)$ at level $n = 2$; $L(17,4)$ (the orbit of $q = J+1$) is balanced, but only because $4^2 \equiv -1 \pmod{17}$, i.e. it is mirror; and $L(17,5)$ (the orbit of $q = 2J-1$) is balanced and chiral. At $J = 2$, $p = 7$, the member $q = J = 2$ of the first family lies in the balanced silver orbit $\{2,3,4,5\}$ (Proposition 7.12) and is balanced too, whereas $L(17,3)$ above is a genuine regularization-only zero. Similarly $\eta_2(L(35,q)) = 0$ for $q \in \{8, 13, 22, 27\}$ with $(N_2^+, N_2^-) = (2,0)$ at $n = 3$ on $L(35,8)$. Balance is rank-specific: $L(17,5)$ is balanced at $J = 3$ but has $(N_2^+, N_2^-) = (2,0)$ at $n = 1$.

Example 4.6 (Small balanced chiral spaces)**.** For $p \le 260$ and $J \le 8$, the chiral rank-J balanced spaces are exactly the following (exact computation; the rows $J \ge 5$ were predicted from Theorem 4.1's families before being computed, see Appendix A):

J	balanced chiral $(p:\ q)$, $p \le 260$
2	7:{2, 3, 4, 5}; 14:{3, 5, 9, 11}
3	17:{5, 7, 10, 12}; 34:{5, 7, 27, 29}
4	31:{7, 9, 22, 24}; 62:{7, 9, 53, 55}
5	7:{2, 3, 4, 5}; 14:{3, 5, 9, 11}; 49:{9, 11, 38, 40}; 98:{9, 11, 87, 89}
6	71:{11, 13, 58, 60}; 142:{11, 13, 129, 131}
7	7:{2, 3, 4, 5}; 14:{3, 5, 9, 11} (blind rank); 97:{13, 15, 82, 84}; 194:{13, 15, 179, 181}
8	127:{15, 17, 110, 112}; 254:{15, 17, 237, 239}

At $J = 5$ the divisor 7 of $4 \cdot 25 - 2 = 98$ enters family (b); at $J = 7$ the spaces $L(7,q)$, $L(14,q)$ enter through the blind rank $7 \mid 2J$. Rank scans at fixed space: $L(7,3)$ and $L(14,3)$ are balanced exactly at $J \in \{2, 5, 7, 9, 12, 14, 16\}$ for $J \le 16$; $L(17,5)$ at $\{3, 14, 17, 20, 31, 34\}$ for $J \le 36$; $L(11,2)$ at $\{11, 22\}$ for $J \le 24$; $L(9,2)$ at no rank (it fails (C)); $L(23,4)$ at $\{9, 14, 23\}$ for $J \le 25$; $L(31,7)$ at $\{4, 27, 31\}$ for $J \le 31$; $L(49,9)$ at $\{5, 44, 49\}$ for $J \le 50$.

5 Divisor decomposition of the eta invariant

Level-wise balance is a strictly smaller phenomenon than $\eta_J = 0$ (Example 4.5). To study the zeros of η_J themselves we first localize $p\,\eta_J$ over the divisors of p.

5.1 Per-element regularized sums

For a nonidentity element g^k of $L(p,q)$ let

$$\eta_k(p,q,J) = \sum_{n \ge 0}^{\mathrm{reg}} E_k(n)\, \nu^{-s}\Big|_{s=0}, \qquad \nu = n + J + 1, \tag{15}$$

the zeta-regularized level sum of the signed character (9). Since $E_k(n)$ is, for fixed k, a finite linear combination of $\nu^j \omega^\nu$ with $j \le 1$ and ω a $2p$th root of unity, its Dirichlet series is a finite combination

of Hurwitz zeta functions, is regular at $s = 0$, and equals there the Abel value of the series [1, Lem. 5.4]; the identity element contributes zero at every level, and $\omega = 1$ never occurs for $k \neq 0$. Hence

$$p\,\eta_J(L(p,q)) = \sum_{k=1}^{p-1} \eta_k(p,q,J). \tag{16}$$

Writing $\theta = \theta_k$, $N = J+1$, and

$$S_{\sin}(a) = \tfrac{1}{2}\cot\tfrac{a}{2} - \sum_{\nu=1}^{N-1}\sin(\nu a), \qquad S_{n\cos}(a) = -\frac{1}{4\sin^2(a/2)} - \sum_{\nu=1}^{N-1}\nu\cos(\nu a)$$

for the Abel values of $\sum_{\nu\geq N}\sin(\nu a)$ and $\sum_{\nu\geq N}\nu\cos(\nu a)$ ($a \notin 2\pi\mathbb{Z}$), the proof of Theorem 3.1 gives the explicit terms

$$\eta_k = \frac{2\big[S_{\sin}(\theta)\sin(Jq\theta) - S_{\sin}(q\theta)\sin(J\theta)\big]}{\cos(q\theta)-\cos\theta} \qquad (k\in\mathcal{N}), \tag{17}$$

$$\eta_k = \pm\frac{2\big[S_{n\cos}(\theta)\sin(J\theta) - J\,S_{\sin}(\theta)\cos(J\theta)\big]}{\sin\theta} \qquad (k\in D^{\pm *}), \tag{18}$$

$$\eta_{p/2} = 0 \qquad (p \text{ even}).$$

Lemma 5.1 (Trace form for nonsingular q)**.** *Let $p \geq 3$ be odd and let the unit q satisfy $\gcd(q-1,p) = \gcd(q+1,p) = 1$ (so every nonidentity element is generic). Then, with $\varphi_k = \pi k/p$,*

$$p\,\eta_J(L(p,q)) = \sum_{k=1}^{p-1} \frac{\cos((2J+1)\varphi_k)\dfrac{\sin(2Jq\varphi_k)}{\sin\varphi_k} - \cos((2J+1)q\varphi_k)\dfrac{\sin(2J\varphi_k)}{\sin(q\varphi_k)}}{\cos(2q\varphi_k)-\cos(2\varphi_k)}, \tag{19}$$

and, after the reindexing $k \mapsto q^* k$ in the second term,

$$p\,\eta_J(L(p,q)) = \sum_{k=1}^{p-1} \mathcal{K}_J(\theta_k)\,\frac{\cos\big((J+\tfrac{1}{2})\theta_k\big)}{\sin(\theta_k/2)}. \tag{20}$$

Proof. Sum the formal character of [1, Sec. 5.1],

$$D_J^{\mathrm{form}}(m;\ell) = \frac{\sin(2m\beta)\sin(2J\alpha) - \sin(2m\alpha)\sin(2J\beta)}{\sin(\alpha+\beta)\sin(\beta-\alpha)}, \qquad \alpha = \frac{\pi\ell q}{p}, \quad \beta = \frac{\pi\ell}{p},$$

over the physical modes $m \geq J+1$ using the Abel value $\sum_{m\geq J+1}\sin(2mx) = \cos((2J+1)x)/(2\sin x)$ and $\sin(\alpha+\beta)\sin(\beta-\alpha) = \frac{1}{2}(\cos 2\alpha - \cos 2\beta)$. Summing over $m \geq J+1$ directly removes the sub-threshold modes, so no L_J term appears. With $\varphi = \beta$ and $\alpha = q\varphi$ this is (19), which is (17) summed over k. The reindexing is Step 3 of Theorem 3.1; in terms of $x = \mathrm{e}^{i\theta}$ the summand of (20) is $f(x) = C(x)[A_q(x) + A_{q^*}(x)]$ with

$$C(x) = \frac{x^{J+1} + x^{-J}}{x-1}, \qquad A_r(x) = \frac{x^{Jr} - x^{-Jr}}{x^r + x^{-r} - x - x^{-1}}, \tag{21}$$

which is even under $x \mapsto 1/x$. □

5.2 Primitive parts

Theorem 5.2 (Divisor decomposition)**.** *For $p \geq 2$, every unit q modulo p and $J \geq 1$,*

$$p\,\eta_J(L(p,q)) = \sum_{\substack{d \mid p \\ d>1}} P_d(q \bmod d,\ J), \qquad P_d(q,J) = \sum_{k \in (\mathbb{Z}/d)^\times} \eta_k(d,q,J), \tag{22}$$

where P_d, the primitive part *of order d, is the regularized contribution of the elements of exact order d and depends only on $(d,\ q \bmod d,\ J)$. Möbius inversion recovers the primitive parts from the eta invariants of the divisor lens spaces:*

$$P_d(q,J) = \sum_{\substack{e \mid d \\ e>1}} \mu(d/e)\, e\, \eta_J(L(e, q \bmod e)). \tag{23}$$

Proof. Group the nonidentity elements in (16) by their exact order d. An element of order d in $L(p,q)$, $k = (p/d)k'$ with $k' \in (\mathbb{Z}/d)^\times$, acts on S^3 as the rotation $(\mathrm{e}^{2\pi i k'/d}, \mathrm{e}^{2\pi i q k'/d})$, which is the element $g^{k'}$ of $L(d, q \bmod d)$; its signed character at every level, hence its regularized sum, depends only on $(d, q \bmod d, J, k')$ (the character products are lift-independent by [1, Lem. 5.2]). The finite sum over group elements commutes with the continuation, and the identity contributes nothing ($H_1 = 0$). Applying (22) to each divisor e and inverting gives (23). □

Corollary 5.3. *(a) At a prime p, $p\,\eta_J(L(p,q)) = P_p(q,J)$ is a single primitive part.*

(b) If $L(p,q)$ is rank-J balanced or mirror, then $P_d(q \bmod d, J) = 0$ for every $d \mid p$.

(c) If every primitive part vanishes then $\eta_J = 0$. The converse fails at composite p: for $(p,q,J) = (42, 11, 9)$ one has $\eta_9 = 0$ with $D_9(0) = 2$, and the primitive parts are $P_3 = 12$, $P_6 = 36$, $P_{21} = -54$, $P_{42} = 6$, $P_2 = P_7 = P_{14} = 0$.

Proof. (a) is the case of a single nontrivial divisor. (b) Balance and the mirror condition descend to every divisor (Corollary 3.4(a); $q^2 \equiv -1 \pmod d$), where $\eta_J = 0$, so every Möbius sum (23) vanishes. (c) The first claim is (22); the example is an exact computation from (6) and (23). □

Proposition 5.4 (Shift invariance and the first-moment form)**.** *Fix p, q, J and write $D_J(r + pk) = v_r + k u_r$ as in (13). For any $c > 0$ the value at $s = 0$ of $\sum_{n \geq 0} D_J(n)(n+c)^{-s}$ is*

$$\sum_{r=0}^{p-1} \Big[v_r \Big(\tfrac{1}{2} - a_r \Big) + u_r \Big(\tfrac{a_r^2}{2} - \tfrac{1}{12} \Big) \Big], \qquad a_r = \frac{r+c}{p},$$

and it is independent of c. In particular, under (C) (where $u_r = 0$ and D_J is p-periodic with $\sum_r v_r = 0$),

$$\eta_J(L(p,q)) = -\frac{1}{p} \sum_{r=0}^{p-1} r\, D_J(r). \tag{24}$$

Proof. The Hurwitz evaluation is [1, Prop. 4.1] with period p. By (12), $pD_J(n) = \nu F(\nu) + G(\nu)$ where F and G are finite Fourier sums over nonidentity elements, so both have mean zero over a period: $\sum_r u_r = 0$ and $p \sum_r v_r = \sum_r (r + J + 1) u_r$. Shifting $c \mapsto c + h$ changes the value by $\frac{h}{p^2}\big[-p \sum_r v_r + \sum_r (r+c) u_r\big] + \frac{h^2}{2p^2} \sum_r u_r$, which vanishes by these two identities. Under (C), $F = 0$, so $u_r = 0$, and $\sum_r v_r = 0$ gives (24). □

Thus at a prime p with $q \not\equiv \pm 1$, $\eta_J = 0$ requires only a vanishing first moment of the periodic sequence $D_J(0), \ldots, D_J(p-1)$, whereas balance requires every term to vanish. This is the algebraic reason why the classification of Section 4 does not classify the zeros of η_J.

5.3 Observations at prime moduli

The following facts were found by exact computation and are recorded as observations; the theorems that explain them are in Sections 6 and 7.

Observation 5.5. Over all primes $41 \le p \le 199$, all units q and $1 \le J \le 12$ (47,952 rows), and all primes $41 \le p \le 101$, all q and $13 \le J \le (p-1)/2$ (23,278 rows), $\eta_J(L(p,q))$ has 1,004 zeros. Every zero is one of: a mirror space; a member of the silver family (b); a member of the four families of [1, Thm. 7.1] at $p = 2J'^2 - 1$ with $J \equiv \pm J'$; or one of 196 further zeros in 49 orbits under $q \mapsto \pm q^{\pm 1}$. Every one of these 49 orbits lies at a silver rank $2J^2 \equiv 1 \pmod p$; the nine silver pairs (p, J) in range are $(41, 12)$, $(71, 6)$, $(97, 7)$, $(127, 8)$, $(199, 10)$, $(47, 20)$, $(73, 16)$, $(79, 35)$, $(89, 32)$, and every one of them carries such zeros. Among them, $q \equiv J$ is a zero at every silver pair in range (the first family of [1], at primes dividing $2J^2 - 1$ rather than equal to it), whereas the analogous extensions of $q = J + 1$ and $q = J(J - 1)$ fail; at $(97, 7)$ and $(199, 10)$ the cube roots of unity modulo p are also zeros. At silver ranks every value in range is an even integer; at other ranks η_J is in general a non-integral rational. For example $\eta_{12}(L(41, 12)) = 0$ with $D_{12}(3) = -2$, while $\eta_{12}(L(41, 1)) = -76$.

Observation 5.6. On primes never inspected before the following three statements were formulated: for every prime $211 \le p \le 331$, every q and $J \le 12$ off the silver ranks (67,032 rows) the 238 zeros are exactly the mirror rows; for every non-silver rank $13 \le J \le (p-1)/2$ at $p \in \{211, 223\}$ (41,286 rows) there is no zero, as both primes are $\equiv 3 \pmod 4$; at the silver rank of each of the 31 primes $p \equiv \pm 1 \pmod 8$ in $[211, 599]$ and every q (12,210 rows) every value is an even integer, and $q = J$ is a zero at all 31 primes. At the adjacent rank $J + 1$, the control computation on the first eight of these primes ($223 \le p \le 281$) has 1,992 nonintegral values among 2,000 rows; only the eight mirror rows are integral, as Corollary 6.4(a) explains. At the silver rank J, the residue $q = J + 1$ fails to be a zero at 22 of the 31 primes. Descriptively, at these silver ranks 471 of 3,076 q-orbits (15.3%) are zeros; the zero fraction is highest for small silver ranks (33% at $p = 241$, $J = 11$; 27% at $p = 449$, $J = 15$; 25% at $p = 337$, $J = 13$), and the extreme values occur at $q = \pm 1$ (up to $|\eta| = 15{,}640$ at $p = 569$).

Observation 5.7 (Composite moduli)**.** Over composites $41 \le p \le 120$, all units q and $J \le 10$ (25,180 rows), the zeros of η_J fall into the classes: balanced (Theorem 4.1) 196; the families of [1, Thm. 7.1] 12; all primitive parts zero but neither of the above 24; and cross-cancellations of 2, 3, 4 nonzero primitive parts, 44, 16, 28 respectively.

6 Congruences for the eta invariant

6.1 Two elementary inputs valid at every modulus

Lemma 6.1 (Low-mode parity)**.** *For every $p \ge 2$, every unit q and all $a, b \ge 1$ with $a \equiv b \pmod 2$, $I_{a,b}(p, q) \equiv \mathbf{1}_{a,b \text{ both odd}} \pmod 2$. Hence $I_{a,b} - I_{b,a}$ is even and $L_J(p, q)$ is even.*

Proof. The involution $(r, t) \mapsto (-r, -t)$ of $W_{a-1} \times W_{b-1}$ preserves the congruence $(1+q)r+(1-q)t \equiv 0 \pmod{2p}$. Its only fixed point is $(0, 0)$, which lies in $W_{a-1} \times W_{b-1}$ exactly when a and b are both odd. □

Lemma 6.2 (Dedekind-sum facts at every modulus)**.** *Let $p \ge 2$ and $1 \le q < p$ with $\gcd(q, p) = 1$. For $1 \le k < p$ put $r_k = \langle qk \rangle_p \in \{1, \dots, p-1\}$, $n_k = \lfloor qk/p \rfloor$, and*

$$R = \sum_{k=1}^{p-1} k\, r_k, \qquad S_2 = \sum_{k=1}^{p-1} k^2 = \frac{p(p-1)(2p-1)}{6}, \qquad \Lambda = \sum_{k=1}^{p-1} n_k^2.$$

Then

$$2qR = (q^2+1)S_2 - p^2\Lambda, \tag{25}$$

$$s(q,p) = \frac{R}{p^2} - \frac{p-1}{4}, \qquad T := 12p\,s(q,p) = \frac{12R}{p} - 3p(p-1) \in 2\mathbb{Z}, \tag{26}$$

$$qT \equiv q^2+1 \pmod{\gcd(3,p)\,p}, \qquad T \equiv q+q^* \pmod{p}. \tag{27}$$

If $p \geq 5$ is prime, moreover $p \mid R$ and $2p\,s(q,p) \in \mathbb{Z}$.

Proof. The residues r_k permute $\{1,\dots,p-1\}$. Expanding $r_k = qk - pn_k$ in $\sum r_k^2 = S_2$ and using $\sum k\,n_k = (qS_2 - R)/p$ gives (25). Expanding the sawtooth definition of $s(q,p)$ gives the first part of (26). Multiplying (25) by 3 shows $p \mid 6qR$, since $(q^2+1)(p-1)(2p-1)/2$ is an integer ($p-1$ is even for odd p, and q^2+1 is even for even p); as $\gcd(q,p)=1$, $p \mid 6R$, so T is an integer, and it is even because $12R/p = 2\cdot(6R/p)$ and $3p(p-1)$ are even. Multiplying (25) by 6, dividing by p and substituting (26) gives the exact identity

$$qT - (q^2+1) = p\big[(q^2+1)(2p-3) - 6\Lambda - 3q(p-1)\big], \tag{28}$$

whose bracket is divisible by 3 when $3 \mid p$; this is (27), the second congruence following on multiplication by q^*. If $p \geq 5$ is prime, 6 is invertible modulo p and $R \equiv qS_2 \equiv 0 \pmod{p}$ (a permutation of the residues changes nothing modulo p), whence $2ps = 2R/p - p(p-1)/2 \in \mathbb{Z}$. □

The congruence $12pq\,s(q,p) \equiv q^2+1 \pmod{\gcd(3,p)p}$ is Rademacher's [24, Ch. 3]; the finite-sum proof above is included so that nothing below rests on an unverified citation.

6.2 Prime moduli

Theorem 6.3 (Prime-modulus congruence)**.** *Let $p \geq 5$ be prime, q a unit modulo p and $J \geq 1$. Then*

(i) $p\,\eta_J(L(p,q)) \in 2\mathbb{Z}$;

(ii) $$p\,\eta_J(L(p,q)) \equiv -\frac{J(2J^2-1)(q+q^{-1})}{3} \pmod{p}.$$

Equivalently, $\eta_J \equiv J(2J^2-1)\,\eta_1$ modulo $\mathbb{Z}_{(p)}$, where $\eta_1 = -4s(q,p)$ and $p\eta_1 = -T/3 \equiv -(q+q^{-1})/3 \pmod{p}$ by Lemma 6.2.

Proof. Multiply (6) by p. The Dedekind term becomes $-2J\cdot(2ps)$, an even integer by Lemma 6.2; the residue term is $2\sum_a (J-a)[\rho_p(aq)+\rho_p(aq^*)]$, visibly even; the aliasing term is $2p$ times an integer; and $-pL_J$ is even by Lemma 6.1. This is (i). Modulo p the last two terms vanish. Since $\rho_p(k) \equiv -2k \pmod{p}$ in both cases of its definition, and $\sum_{a=1}^{J-1} a(J-a) = J(J^2-1)/6$,

$$2\sum_{a=1}^{J-1}(J-a)\big[\rho_p(aq)+\rho_p(aq^*)\big] \equiv -4(q+q^*)\sum_{a=1}^{J-1} a(J-a) = -\tfrac{2}{3}J(J^2-1)(q+q^*),$$

while $-4Jps = -\frac{J}{3}T \equiv -\frac{J}{3}(q+q^*)$ by (27); here 3 is invertible because $p \geq 5$. Adding, $p\,\eta_J \equiv -\frac{J}{3}(q+q^*)[1+2(J^2-1)] = -J(2J^2-1)(q+q^*)/3$, which is (ii). □

Corollary 6.4 (Integrality, denominators, and the zero set off the silver ranks)**.** *Let $p \geq 5$ be prime, q a unit and $J \geq 1$.*

(a) $\eta_J(L(p,q)) \in \mathbb{Z}$ *iff* $p \mid J(2J^2-1)(q^2+1)$, *and then* $\eta_J \in 2\mathbb{Z}$. *Otherwise the reduced denominator of* η_J *is exactly* p; *in particular* $\eta_J \neq 0$.

(b) (Silver integrality.) *If* $2J^2 \equiv 1 \pmod p$ *then* $\eta_J(L(p,q)) \in 2\mathbb{Z}$ *for every unit* q.

(c) (Dichotomy off the silver ranks.) *If* $2J^2 \not\equiv 1 \pmod p$ *and* $p \nmid J$, *then* $\eta_J(L(p,q)) = 0$ *iff* $q^2 \equiv -1 \pmod p$.

(d) (Blind ranks.) *If* $p \mid J$ *then* $\eta_J(L(p,q)) = 0$ *for every* $q \not\equiv \pm 1$, *while for* $J = np$, $n \geq 1$,

$$\eta_{np}(L(p,\pm 1)) = \mp \frac{n(p^2-1)}{6} \neq 0, \tag{29}$$

and $\eta_{J+p}(L(p,\pm 1)) - \eta_J(L(p,\pm 1)) = \mp(p^2-1)/6$ *for every* $J \geq 1$.

Consequently, for a prime $p \geq 5$ *and a rank* J *with* $2J^2 \not\equiv 1 \pmod p$, *the zero set of* $\eta_J(L(p,\cdot))$ *is completely known: the mirror spaces if* $p \nmid J$, *and all* $q \not\equiv \pm 1$ *if* $p \mid J$. *At* $J = 1$, *(c) is the classical Dedekind zero criterion at prime moduli.*

Proof. (a) Write $p\,\eta_J = 2\Theta$ with $\Theta \in \mathbb{Z}$ by (i). Then $\eta_J \in \mathbb{Z}$ iff $p \mid 2\Theta$ iff $p \mid \Theta$ (p odd), and then $\eta_J = 2\Theta/p \in 2\mathbb{Z}$; otherwise the denominator is exactly p. By (ii), $p \mid \Theta$ iff $p \mid J(2J^2-1)(q+q^*)$ iff $p \mid J(2J^2-1)(q^2+1)$ (multiply by the unit q). (b) is (a) with $p \mid 2J^2-1$. (c) Under the hypotheses $p \nmid J(2J^2-1)$, so by (a) η_J is nonzero unless $p \mid q^2+1$; conversely mirror spaces have $\eta_J = 0$ by (5). (d) The vanishing for $q \not\equiv \pm 1$ is the rank reflection $\eta_p = 0$ and periodicity of [1, Thm. 6.2], whose hypothesis $\gcd(1 \pm q, p) = 1$ holds at a prime for $q \not\equiv \pm 1$. For $q = 1$ every element is a Clifford translation and, by (18) and (21), in the variable $x = \mathrm{e}^{i\theta_k}$,

$$\eta_k = f_{J,1}(x) := \frac{J\,C(x)\,(x^J + x^{-J}) - C'(x)\,(x^J - x^{-J})}{x - x^{-1}}, \qquad C'(x) = x\frac{dC}{dx},$$

so that $p\,\eta_J(L(p,1)) = \mathrm{Tr}_{\mathbb{Q}(\zeta_p)/\mathbb{Q}}\, f_{J,1}(\zeta_p)$. At $J = np$ and a primitive pth root of unity, $x^J = x^{-J} = 1$ and $x^{J+1} = x$, so

$$f_{J,1}(x) = \frac{2J\,C(x)}{x - x^{-1}} = \frac{2J(x+1)}{(x-1)(x-x^{-1})} = \frac{2Jx}{(x-1)^2} = 2J(\lambda^{-2} - \lambda^{-1}), \qquad \lambda = 1 - x.$$

The elementary traces $\mathrm{Tr}\,\lambda^{-1} = (p-1)/2$ and $\mathrm{Tr}\,\lambda^{-2} = -(p-1)(p-5)/12$ (obtained by differentiating $\Phi_p(x) = 1 + \cdots + x^{p-1}$ at $x = 1$) give $\mathrm{Tr}\, f_{J,1} = -J(p^2-1)/6$, hence (29) at $q = 1$; the case $q = -1$ is the sign (5). For the rank increment, $C_{J+p} = C_J$ at pth roots of unity while $C'_{J+p} - C'_J = p(x^{J+1} - x^{-J})/(x-1)$, and substitution gives $f_{J+p,1} - f_{J,1} = 2px/(x-1)^2$, whose trace is $-p(p^2-1)/6$. □

Remark 6.5 (Cyclotomic-trace route)**.** Theorem 6.3 was first proved by a different route, which we sketch because it explains the shape of the congruence. By Lemma 5.1, for $q \not\equiv \pm 1$ one has $p\,\eta_J = \mathrm{Tr}_{K/\mathbb{Q}}\, f(\zeta)$ with $K = \mathbb{Q}(\zeta)$, $\zeta = \mathrm{e}^{2\pi i/p}$, $f = C\,[A_q + A_{\bar q}]$ as in (21) ($\bar q$ a positive integer inverse of q), and $p\,\eta_J = \pm\,\mathrm{Tr}\, f_{J,1}(\zeta)$ for $q \equiv \pm 1$. The denominators factor as $x^r + x^{-r} - x - x^{-1} = x^{-r}(x^{r-1}-1)(x^{r+1}-1)$, so f is integral away from p, and at $\lambda = 1 - \zeta$ it has a pole of order at most two with p-integral Laurent coefficients: $f(\zeta) = c_{-2}\lambda^{-2} + c_{-1}\lambda^{-1} + c_0 + \lambda h$, $h \in \mathbb{Z}_{(p)}[\zeta]$. Since $\mathrm{Tr}(\lambda h) \in p\mathbb{Z}_{(p)}$ and the traces of $\lambda^{-1}, \lambda^{-2}, 1$ are congruent modulo p to $-\frac{1}{2}, -\frac{5}{12}, -1$, which are minus the constant terms of $(1-\mathrm{e}^u)^{-1}, (1-\mathrm{e}^u)^{-2}, 1$, one gets $\mathrm{Tr}\, g(\zeta) \equiv -[u^0]g(\mathrm{e}^u) \pmod p$ for such g. The constant term of $f(\mathrm{e}^u)$ is $4J\sum_{r\in\{q,\bar q\}} \frac{r}{r^2-1}\big[\frac{J^2}{2} + \frac{J}{2} + \frac{J^2r^2}{6} - \frac{r^2}{12}\big]$, which after $\bar q \equiv q^{-1}$ collapses to $J(2J^2-1)(q^2+1)/(3q)$; this is (ii). Evenness follows because f is even under $x \mapsto 1/x$, so $\mathrm{Tr}_{K/\mathbb{Q}}\, f = 2\,\mathrm{Tr}_{K^+/\mathbb{Q}}\, f$, and f lies in the inverse different of $K^+ = \mathbb{Q}(\zeta + \zeta^{-1})$ (valuation ≥ -1 at the prime above p, against the exponent $-(p-3)/2$ for $p \geq 5$).

Remark 6.6. Even integrality never by itself gives a zero: $p = 7$, $J = 2$, $q = 1$ is a silver row with $\eta_2(L(7,1)) = -2$. The boundary $p \geq 5$ is substantive: $\eta_1(L(3,1)) = -2/9$, so (i) fails at $p = 3$, and the division by 3 in (ii) is unavailable there.

6.3 Every modulus

Theorem 6.7 (Universal congruence)**.** *For every modulus $p \geq 2$, every unit q modulo p and every $J \geq 1$,*

$$\eta_J(L(p,q)) - J(2J^2-1)\,\eta_1(L(p,q)) \in 2\mathbb{Z}, \qquad \eta_1(L(p,q)) = -4s(q,p). \tag{30}$$

Proof. Put $N = J(J^2-1)/6 = \sum_{a=1}^{J-1} a(J-a) \in \mathbb{Z}$ and fix representatives $q, q^* \in [1, p-1]$ with $qq^* \equiv 1$. Subtracting $J(2J^2-1)\eta_1 = -4J(2J^2-1)s$ from (6), the Dedekind term becomes $-4Js + 4J(2J^2-1)s = 8J(J^2-1)s = 48Ns$. For the residue term use $\rho_p(k) = p - 2\langle k\rangle_p - p\,\delta_p(k)$ and $\langle aq\rangle_p = aq - p\lfloor aq/p\rfloor$:

$$\frac{2}{p}\sum_{a=1}^{J-1}(J-a)\big[\rho_p(aq) + \rho_p(aq^*)\big] = -\frac{4N}{p}(q+q^*) + (\text{even integers}).$$

The aliasing double sum in (6) is $2\times$ an integer, and L_J is even by Lemma 6.1. Hence

$$\eta_J - J(2J^2-1)\eta_1 = \frac{4N}{p}\big(12ps - (q+q^*)\big) + (\text{even integers}) - L_J.$$

By (27), $12ps - (q+q^*) = T - (q+q^*) \in p\mathbb{Z}$, so the first term lies in $4N\mathbb{Z} \subseteq 2\mathbb{Z}$. □

Theorem 6.8 (Exact denominator and necessary conditions for a zero)**.** *Let $p \geq 2$, q a unit, $J \geq 1$, and put $C = J(2J^2-1)$ and $T = 12p\,s(q,p) \in 2\mathbb{Z}$. Then*

(a) the reduced denominator of $\eta_J(L(p,q))$ is $3p/\gcd(3p, CT)$; in particular $\eta_J \in \mathbb{Z}$ iff $3p \mid CT$, and $\eta_J \in 2\mathbb{Z}$ iff $6p \mid CT$;

(b) if $\eta_J(L(p,q)) = 0$ then $p \mid C(q^2+1)$; if moreover $4 \mid p$ then $2^{v_2(p)} \mid J$, and if $3 \mid p$ then $3^{v_3(p)+1} \mid J$;

(c) writing $p = 2^a 3^b m$ with $\gcd(m, 6) = 1$,

$$\eta_J(L(p,q)) \in 2\mathbb{Z} \iff m \mid C(q^2+1) \quad \text{and} \quad (a < 2 \text{ or } 2^a \mid J) \quad \text{and} \quad (b = 0 \text{ or } 3^{b+1} \mid J);$$

(d) if $3 \nmid p$, then $p\,\eta_J \in 2\mathbb{Z}$ and $p\,\eta_J \equiv -C(q+q^*)/3 \pmod p$; that is, both parts of Theorem 6.3 hold at every modulus coprime to 3.

Proof. By Theorem 6.7 and $\eta_1 = -T/(3p)$ there is $H \in \mathbb{Z}$ with $\eta_J = -CT/(3p) + 2H$; adding an integer does not change a reduced denominator, which proves (a). For (b), a zero gives $3p \mid CT$, hence $p \mid CT$, and by (27) $C(q^2+1) \equiv qCT \equiv 0 \pmod p$. If $4 \mid p$ then q is odd, $q^2+1 \equiv 2 \pmod 4$, and (27) modulo 4 gives $v_2(T) = 1$; a zero needs $6p \mid CT$, so $v_2(C) \geq v_2(p)$, and $v_2(C) = v_2(J)$ since $2J^2-1$ is odd. If $3 \mid p$ then (27) modulo 3 gives $qT \equiv q^2+1 \equiv 2$, so $v_3(T) = 0$, and $6p \mid CT$ forces $v_3(C) \geq v_3(p) + 1$; $v_3(C) = v_3(J)$ because $2J^2 - 1 \not\equiv 0 \pmod 3$. For (c), the m-part: $T \equiv q + q^* \pmod m$, so $m \mid CT$ iff $m \mid C(q+q^*)$ iff $m \mid C(q^2+1)$. The 2-part: T is always even; if $a = 1$, write $p = 2u$ with u odd: $r_k \equiv k \pmod 2$ so $R \equiv \sum k = u(2u-1)$ is odd, $u \mid 3R$ makes $3R/u$ odd, and $T = 2[3R/u - 3u(2u-1)]$ is divisible by 4; if $a \geq 2$, $v_2(T) = 1$ as above. The 3-part: if $b = 0$ then $p \mid 6R$ and $\gcd(p, 3) = 1$ give $p \mid 2R$, so $3 \mid T$; if $b \geq 1$, $v_3(T) = 0$. Comparing with $v_2(6p) = 1 + a$ and $v_3(6p) = 1 + b$ gives (c). For (d), if $3 \nmid p$ then $T \in 6\mathbb{Z}$ by the 3-part just shown, so $CT/3 \in 2\mathbb{Z}$ and $p\,\eta_J = -CT/3 + 2pH \in 2\mathbb{Z}$; reducing modulo p and using $T \equiv q + q^*$ with 3 invertible gives the congruence. □

Remark 6.9 (Boundaries)**.** The conditions in (b) are necessary, not sufficient: $\eta_{25}(L(25,1)) = -104$ although $25 \mid C(q^2+1)$ with $C = 25(2\cdot 625 - 1)$. At even moduli an integral value need not be even: $\eta_2(L(4,1)) = -1$, $\eta_1(L(4,1)) = -1/2$, and the difference (30) is 6. At $p = 2$ the two branch multiplicities agree at every level, so $\eta_J(L(2,1)) = 0$ for all J and (d) holds trivially. Prime powers must be read with valuations: at $p = 25$, $J = 5$, $q = 2$ one factor 5 comes from J and one from q^2+1. Part (b) with the full modulus replaced by its largest divisor coprime to 6 is the weaker statement one obtains by localizing at primes $\ell \geq 5$.

Remark 6.10 (Status of the zero problem after this section)**.** For prime $p \geq 5$ the zero set of $\eta_J(L(p,\cdot))$ is known at every rank with $2J^2 \not\equiv 1 \pmod p$ (Corollary 6.4). At a silver rank it is an even integer whose zeros include the balanced family (b), the family of Theorem 7.1 below, mirror spaces when $p \equiv 1 \pmod 4$, and further zeros not yet characterized (Section 9). At composite moduli Theorem 6.8 gives necessary local conditions only; Observation 5.7 shows that cross-cancellations between primitive parts occur. Congruences of the type of Theorems 6.3 and 6.7 may follow from index theory for flat bundles; that literature has not been searched, and no novelty is claimed for them.

7 The silver zero at $q = J$

7.1 Statement

Theorem 7.1. *For every $J \geq 2$ and every divisor $p > 1$ of $2J^2 - 1$,*

$$\eta_J(L(p,J)) = 0,$$

and hence also $\eta_J(L(p,q)) = 0$ for $q \equiv \pm J$ and $q \equiv \pm 2J \pmod p$.

For $p = 2J^2 - 1$ this is the first family of [1, Thm. 7.1], proved there by repeated Dedekind reciprocity with residues that do not wrap modulo p. At a proper divisor the residues wrap and that argument does not apply; the proof below uses instead the conjugacy invariance of Rademacher's symbol. The transfer to $q \equiv \pm J, \pm 2J$ is (5) together with $2J \equiv J^{-1} \pmod p$.

Throughout, fix $J \geq 2$ and a divisor $p > 1$ of $N := 2J^2 - 1$. Then p is odd, coprime to 3, and $m_0 := N/p$ is odd. The residues $2J$, $J \pm 1$, $2J \pm 1$ are units modulo p: a prime dividing $J \pm 1$ and N would divide $2J^2 - 1 - 2(J \mp 1)(J \pm 1) = 1$; $(2J+1)(2J-1) = 2N + 1 \equiv 1$; and $(J+1)(J-1) \equiv -\frac{1}{2}$. In particular $q = J$ is nonsingular in the sense of Lemma 5.1: no nonidentity element is a Clifford translation, and this holds at composite p as well. Put

$$\varepsilon := 4J + 3, \qquad \lambda := -\varepsilon, \qquad \lambda^2 + 6\lambda + 1 = \varepsilon^2 - 6\varepsilon + 1 = 8N \equiv 0 \pmod p; \tag{31}$$

ε is the silver unit $3 + 2\sqrt{2}$ read modulo p with $\sqrt{2} \equiv 2J$. Two notational conventions are essential: $c_m(z)$ denotes the centered residue of z modulo m in $(-m/2, m/2]$; and a fraction a/b written inside $c_p(\cdot)$ means the modular quotient ab^{-1} in $\mathbb{Z}/p$, not a real number. For instance $c_p(j + \frac{1}{2}) = j - \frac{p-1}{2}$ for $0 \leq j < p$, and $c_{17}(\frac{1}{2}) = c_{17}(9) = -8$. Finally $s_n(a,b;p)$ is the Fourier–Dedekind sum [10]

$$s_n(a,b;p) = \frac{1}{p} \sum_{\substack{\zeta^p = 1 \\ \zeta \neq 1}} \frac{\zeta^n}{(\zeta^a - 1)(\zeta^b - 1)}, \qquad \gcd(a,p) = \gcd(b,p) = 1,$$

which satisfies $s_n = s_{a+b-n}$ and $s_n(a,-b) = -s_{n+b}(a,b)$.

7.2 Reduction to a quarter-shifted covariance

Lemma 7.2 (Step 1: a trigonometric sum)**.** *With $\varphi_k = \pi k/p$,*

$$p\,\eta_J(L(p,J)) = \sum_{k=1}^{p-1} \frac{(-1)^k \cos(\pi k/p)}{\cos(2\pi k/p) - \cos(2\pi Jk/p)}.$$

Proof. Apply (19) with $q = J$. Write $2J^2 = 1 + m_0 p$ with m_0 odd; then $\sin(2J^2\varphi_k) = (-1)^k \sin\varphi_k$, $\cos((2J+1)J\varphi_k) = (-1)^k \cos((J+1)\varphi_k)$, $\sin(2J\varphi_k)/\sin(J\varphi_k) = 2\cos(J\varphi_k)$, and $\cos((2J+1)\varphi) - 2\cos((J+1)\varphi)\cos(J\varphi) = -\cos\varphi$. The numerator of (19) becomes $-(-1)^k \cos\varphi_k$ and the denominator $\cos(2J\varphi_k) - \cos(2\varphi_k)$. □

Lemma 7.3 (Step 2: Fourier–Dedekind form)**.** $p\,\eta_J(L(p,J)) = 2p\,s_J(2J+1,\,2J-1;\,p)$.

Proof. Put $h = (p+1)/2$, $\theta = 2\pi k/p$ and $\zeta = \mathrm{e}^{i\theta}$. Then $(-1)^k\cos(\pi k/p) = \cos(h\theta)$ and $\cos\theta - \cos(J\theta) = -\frac{1}{2}\zeta^{-J}(\zeta^{J+1}-1)(\zeta^{J-1}-1)$, so by Lemma 7.2, using $s_n = s_{a+b-n}$ with $a+b = 2J$, $p\,\eta_J = -2p\,s_{J+h}(J+1, J-1; p)$. The substitution $\zeta \mapsto \zeta^{2J}$ (a permutation of the nontrivial pth roots of unity, as $\gcd(2J,p)=1$) sends the exponents $(J+1, J-1, J+h)$ to $(2J+1, -(2J-1), J+1)$ modulo p, because $2J^2 \equiv 1$ and $2h \equiv 1$. Then $s_n(a,-b) = -s_{n+b}(a,b)$ and once more $s_n = s_{a+b-n}$ give $s_{J+h}(J+1, J-1) = -s_{3J}(2J+1, 2J-1) = -s_J(2J+1, 2J-1)$. □

Lemma 7.4 (Step 3: covariance form)**.** *Let $K := \sum_{x\in\mathbb{Z}/p} c_p(x+\frac14)\,c_p(\lambda x+\frac14)$. Then $\eta_J(L(p,J)) = 2K/p^2$; in particular $\eta_J(L(p,J)) = 0 \iff K = 0$.*

Proof. For $\zeta^p = 1$, $\zeta \neq 1$ and a a unit, $1/(\zeta^a - 1) = \frac{1}{p}\sum_{j=0}^{p-1} j\zeta^{aj}$. Inserting this for both factors of $s_n(a,b;p)$ and using orthogonality of the pth roots of unity gives, with $\mu = (p-1)/2$,

$$p\,s_n(a,b;p) = \frac{1}{p}\sum_{j_1=0}^{p-1}(j_1-\mu)(j_2-\mu), \qquad aj_1 + bj_2 + n \equiv 0 \pmod p,\ 0 \le j_2 < p.$$

Now $j - \mu = c_p(j+\frac12)$. For $(a,b,n) = (2J+1, 2J-1, J)$, using $b^{-1} \equiv a$ and $a^2 \equiv 4J+3 = \varepsilon$, the constraint reads $y_2 \equiv -\varepsilon y_1 + (1+J)$ for $y_i = j_i + \frac12$, and $1 + J = (1+\varepsilon)/4$ in $\mathbb{Z}/p$. With $x = y_1 - \frac14$ this is $y_2 = \lambda x + \frac14$, so $p\,s_J = K/p$, and Lemma 7.3 gives $\eta_J = 2K/p^2$. □

7.3 From the quarter shift to Dedekind sums at even moduli

Lemma 7.5 (Step 4: lifting to modulus $4p$)**.** *Put $u = c_p(x+\frac14)$, $w = c_p(\lambda x+\frac14)$, $U = 4u-1$, $W = 4w-1$. Fix $\hat\lambda \in \mathbb{Z}$ with $\hat\lambda \equiv \lambda \pmod p$ and $\hat\lambda \equiv 1 \pmod 4$. Then $\hat\lambda^2 + 6\hat\lambda + 1 \equiv 0 \pmod{4p}$, and as x runs over $\mathbb{Z}/p$, U runs over the p integers $\equiv 3 \pmod 4$ in $(-2p, 2p)$, with $W = c_{4p}(\hat\lambda U)$. Consequently*

$$16K = T - p, \qquad T := \sum_{\substack{U\in\mathbb{Z}/4p \\ U\equiv 3\,(4)}} c_{4p}(U)\,c_{4p}(\hat\lambda U). \tag{32}$$

Proof. The congruence modulo $4p$ holds modulo p by (31) and modulo 4 because $1+6+1 = 8$. As x runs over $\mathbb{Z}/p$ so does $x+\frac14$, so u runs over the centered residues and $U = 4u-1$ over the stated set; since λ is a unit the same holds for w and W. Now $W \equiv 4\lambda x \equiv \hat\lambda U \pmod p$ and $W \equiv 3 \equiv \hat\lambda U \pmod 4$, so $W \equiv \hat\lambda U \pmod{4p}$, and $W \in (-2p, 2p)$ is the centered representative. Finally $\sum U = \sum W = -p$ (the centered residues sum to zero), so $16K = \sum(U+1)(W+1) = T - p - p + p$. □

Lemma 7.6 (Step 5: parity split)**.** *For an even modulus m and h odd and coprime to m put $D_m(h) := \sum_{V\in\mathbb{Z}/m} c_m(V)\, c_m(hV)$. Then $2T = D_{4p}(\hat\lambda) - 4D_{2p}(\hat\lambda)$.*

Proof. $U \mapsto -U$ exchanges the odd classes 1 and 3 modulo 4 and preserves each summand ($\hat\lambda$ is odd and $\pm 2p$ is even), so the sum over odd U is $2T$. The even $U = 2V$ contribute $c_{4p}(2V)c_{4p}(2\hat\lambda V) = 4c_{2p}(V)c_{2p}(\hat\lambda V)$. □

Lemma 7.7 (Centered residues and Dedekind sums)**.** *For even m and h odd and coprime to m, $D_m(h) = \dfrac{m^2}{4} + m^2\, s(h,m)$.*

Proof. For $V \not\equiv m/2$, $c_m(V)/m = \left(\!\left(V/m + \tfrac12\right)\!\right)$; at $V \equiv m/2$ the left side is $\tfrac12$ and the right side 0, and $hV \equiv m/2$ iff $V \equiv m/2$ because h is odd. Hence $D_m/m^2 = \tfrac14 + \sum_V \left(\!\left(V/m + \tfrac12\right)\!\right)\left(\!\left(hV/m + \tfrac12\right)\!\right)$. Use $\left(\!\left(x + \tfrac12\right)\!\right) = ((2x)) - ((x))$ in both factors. Of the four resulting sums, $\sum_V ((2V/m))\,((2hV/m)) = 2s(h, m/2)$; the two mixed sums each equal $s(h,m/2)$, by splitting V modulo $m/2$ and using $((y)) + \left(\!\left(y + \tfrac12\right)\!\right) = ((2y))$; and $\sum_V ((V/m))\,((hV/m)) = s(h,m)$. The signed total is $2s(h,m/2) - 2s(h,m/2) + s(h,m)$. □

Combining Lemmas 7.5–7.7, $2T = 16p^2\big[s(\hat\lambda, 4p) - s(\hat\lambda, 2p)\big]$, and therefore

$$\eta_J(L(p,J)) = 0 \iff K = 0 \iff T = p \iff s(\hat\lambda,4p) - s(\hat\lambda,2p) = \frac{1}{8p}. \tag{33}$$

7.4 Rademacher's symbol and the centralizer

For $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$ Rademacher's functions are

$$\Phi(M) = \begin{cases} \dfrac{a+d}{c} - 12\ \mathrm{sgn}(c)\, s(d,|c|), & c \neq 0, \\ \dfrac{b}{d}, & c = 0, \end{cases} \qquad \Psi(M) = \Phi(M) - 3\ \mathrm{sgn}\big(c(a+d)\big), \tag{34}$$

and Ψ is invariant under $SL_2(\mathbb{Z})$-conjugation on hyperbolic elements [24, Ch. 4]; this normalization agrees with [12, eq. (1.2)] and [21, eq. (1.2)]. Directly from the definition, with $F = \mathrm{diag}(1,-1)$,

$$\Psi(FMF) = -\Psi(M), \tag{35}$$

since $FMF = \begin{pmatrix} a & -b \\ -c & d \end{pmatrix}$ negates Φ and $\mathrm{sgn}(c(a+d))$.

Lemma 7.8 (Dedekind sums at the silver unit, even moduli)**.** *For $m \in \{2p, 4p\}$, $s(\hat\lambda, m) = \dfrac{m-2}{4m}$.*

Proof. Write $\hat\lambda = 1 + 4t$; then $\hat\lambda^2 + 6\hat\lambda + 1 = 8\sigma$ with $\sigma = 1 + 4t + 2t^2$ odd and $p \mid \sigma$. For $m \in \{2p, 4p\}$ the integer $b := -(\hat\lambda^2 + 6\hat\lambda + 1)/m$ is even ($-4\sigma/p$ or $-2\sigma/p$). Let

$$M = \begin{pmatrix} -6 - \hat\lambda & b \\ m & \hat\lambda \end{pmatrix} \in SL_2(\mathbb{Z}), \qquad \mathrm{tr}\, M = -6,$$

which is hyperbolic. Since $\hat\lambda + 3 \equiv 0 \pmod 4$ and b, m are even, $A := (M + 3I)/2$ is integral; Cayley–Hamilton ($M^2 + 6M + I = 0$) gives $A^2 = 2I$, $\mathrm{tr}\, A = 0$, $\det A = -2$. Hence $R := I + A$ is

integral, commutes with M, and has $\det R = (1+\sqrt{2})(1-\sqrt{2}) = -1$. Put $P = RF$; then $\det P = 1$ and $P^{-1}MP = FR^{-1}MRF = FMF$. By conjugation invariance and (35), $\Psi(M) = \Psi(FMF) = -\Psi(M)$, so $\Psi(M) = 0$. Now $c = m > 0$ and $a + d = -6 < 0$, so

$$0 = \Psi(M) = -\frac{6}{m} - 12\, s(\hat{\lambda}, m) + 3, \qquad s(\hat{\lambda}, m) = \frac{3 - 6/m}{12} = \frac{m-2}{4m}.$$

With the unreduced lift $\hat{\lambda} = -(4J+3)$ (already $\equiv 1 \pmod 4$) and $n_0 = N/p$, the centralizers are explicit: $A_{2p} = \begin{pmatrix} 2J & -2n_0 \\ p & -2J \end{pmatrix}$ and $A_{4p} = \begin{pmatrix} 2J & -n_0 \\ 2p & -2J \end{pmatrix}$, both squaring to $2I$ because $4J^2 - 2pn_0 = 2$. □

Proof of Theorem 7.1. By Lemma 7.8, $s(\hat{\lambda}, 4p) - s(\hat{\lambda}, 2p) = \frac{4p-2}{16p} - \frac{2p-2}{8p} = \frac{1}{8p}$, so (33) gives $K = 0$ and $\eta_J(L(p,J)) = 0$. □

Remark 7.9 (The binary-form route)**.** Lemma 7.8 was first proved through binary quadratic forms. The form $Q_M = [m,\, 2\hat{\lambda}+6,\, -b]$ attached to M (via $M \mapsto [c,\, d-a,\, -b]$) has discriminant 32 and content exactly 2, since $32/\,\text{content}^2$ must be $\equiv 0, 1 \pmod 4$. Its primitive part has discriminant 8; because $\mathbb{Z}[\sqrt{2}]$ is norm-Euclidean, hence a principal ideal domain, and has the unit $1+\sqrt{2}$ of norm -1, there is a single proper class of primitive forms of discriminant 8, so M is $SL_2(\mathbb{Z})$-conjugate to $M_0 = \begin{pmatrix} -3 & 4 \\ 2 & -3 \end{pmatrix}$ (whose form is $[2, 0, -4]$), and $\Psi(M_0) = -3 - 12s(1,2) + 3 = 0$. The SL_2 (rather than GL_2) conjugacy is obtained by adjusting an integral O-basis of $\mathbb{Z}^2$ by $I + A_0$, $A_0 = \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}$, which has determinant -1 and commutes with M_0. The centralizer proof above avoids the class-number input. Both routes rest on the conjugacy invariance of Ψ, which we cite and do not re-prove.

Remark 7.10 (Why a quarter shift, and why silver ranks)**.** The shift $\frac{1}{4}$ in Lemma 7.4 is what forces $\hat{\lambda} \equiv 1 \pmod 4$. With $\hat{\lambda} \equiv 3 \pmod 4$ the form at $m = 4p$ is primitive, $\Psi = \pm 3$, and the evaluation fails (for $(J,p) = (3,17)$ one gets $s = -1/136$ instead of $33/136$). The unit $1+\sqrt{2}$ of norm -1 is what makes the content-two class have $\Psi = 0$; in the language of Section 8, M is *inert*.

7.5 Consequences and clarifications

Corollary 7.11 (The Dedekind sum at the silver unit, odd moduli)**.** *Let p be odd and let $\varepsilon \in \mathbb{Z}/p$ satisfy $\varepsilon^2 - 6\varepsilon + 1 \equiv 0 \pmod p$. Then*

$$s(\varepsilon, p) = \begin{cases} -\dfrac{p-1}{2p}, & p \equiv 1 \pmod 8, \\[2ex] \dfrac{1}{2p}, & p \equiv 7 \pmod 8. \end{cases}$$

Proof. Lift $h = -\varepsilon$ to $h \equiv 1 \pmod 4$ and write $h^2 + 6h + 1 = 8\sigma$ with σ odd and $p \mid \sigma$. The matrix $M_p = \begin{pmatrix} -6-h & -8\sigma/p \\ p & h \end{pmatrix}$ has trace -6, and its form $H = [p,\, 2h+6,\, 8\sigma/p]$ has discriminant 32 and is primitive (p is odd). The integral form $G = [p,\, h+3,\, 2\sigma/p]$ has discriminant 8 and $H(x,y) = G(x, 2y)$; by Remark 7.9 G is properly equivalent to $x^2 - 2y^2$, so H is that norm form restricted to an oriented sublattice of index two. There are three such sublattices, the preimages of the three lines in $(\mathbb{Z}/2)^2$: on $\{x \text{ even}\}$ the norm is always even, impossible since H represents

the odd number p; on $\{y \text{ even}\}$ the restricted form is $x^2 - 8y^2$; on $\{x \equiv y \pmod 2\}$, with the positively oriented basis $(1,1),(0,2)$, it is $-x^2 - 8xy - 8y^2$, properly equivalent to $-x^2 + 8y^2$ by a unimodular substitution. The odd values of these two forms are respectively $\equiv 1$ and $\equiv 7 \pmod 8$, so they are not properly equivalent, and H is determined by $p \bmod 8$. At fixed trace t the map $M \mapsto Q_M = [c, d-a, -b]$ is injective: from $[u,v,w]$ one recovers $c = u$, $b = -w$, $a = (t-v)/2$ and $d = (t+v)/2$. It is equivariant because $Q_M(x) = \det(x, Mx)$ gives $Q_{P^{-1}MP}(x) = Q_M(Px)$ for $P \in SL_2(\mathbb{Z})$. Thus M_p is conjugate to $M_+ = \begin{pmatrix} -3 & 8 \\ 1 & -3 \end{pmatrix}$ ($\Psi = -3$) if $p \equiv 1 \pmod 8$ and to $M_- = \begin{pmatrix} -3 & -8 \\ -1 & -3 \end{pmatrix}$ ($\Psi = +3$) if $p \equiv 7 \pmod 8$. Since $\Psi(M_p) = 3 - 6/p - 12s(h,p) = 3 - 6/p + 12s(\varepsilon, p)$, solving gives the two values. □

Proposition 7.12 (The silver zero and level balance)**.** *Let $J \geq 2$ and $p > 1$ divide $2J^2 - 1$. Then $L(p, J)$ is chiral, and it is rank-J balanced if and only if $p = 7$. Thus for $p \neq 7$ Theorem 7.1 produces zeros of η_J which are not level-wise balanced; for example at $(J, p) = (3, 17)$ the imbalance sequence begins $D_3 = (0, 0, -2, 4, -2, 2, -4, 2, 0, \ldots)$, and only its Hurwitz moments cancel.*

Proof. Mirror balance would require $J^2 \equiv -1$ together with $2J^2 \equiv 1$, forcing $p \mid 3$, but $3 \nmid 2J^2 - 1$. A blind rank is impossible since J is a unit. For the silver family, $J \equiv 2J \pm 1$ forces $J \equiv \mp 1$, contradicting $p \mid 2J^2 - 1$; $J \equiv -(2J \pm 1)$ gives $3J \equiv \mp 1$, hence $9J^2 \equiv 1$ and, with $2J^2 \equiv 1$, $9 \equiv 2$, i.e. $p \mid 7$. Conversely at $p = 7$, $2J^2 \equiv 1$ gives $J \equiv \pm 2$, and $q = J$ then lies in $\{\pm(2J \pm 1)\} = \{2, 3, 4, 5\}$. The sequence quoted is an exact computation. □

Remark 7.13. The other universal zero orbit at silver ranks, $q \equiv \pm(2J \pm 1)$, is the balanced family (b) of Theorem 4.1; in (19) its two terms are exchanged with opposite signs by $k \mapsto (2J+1)k$, which gives a second, Fourier-side proof of $\eta_J = 0$ for that family. Theorems 6.3 and 6.7 are not used in the proof of Theorem 7.1; they are consistent with it, an exact vanishing being finer than any congruence.

8 Zero versus symmetry: the Rademacher symbol

The proof of Lemma 7.8 turned on an integral matrix of determinant -1 commuting with a hyperbolic $M \in SL_2(\mathbb{Z})$, which forces $\Psi(M) = 0$. This section records, with citations, the extent to which that mechanism is necessary. None of the facts about Ψ below is claimed as new: the forward implications are classical, and the negative answer also follows from known asymptotic counts. What we add are explicit small witnesses with elementary proofs, exact statements for the two- and four-block classes, an elementary density bound, and the identification of the matrices behind Theorem 7.1.

8.1 Conventions and cited inputs

Let $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$ be hyperbolic with $t = a + d$, and Φ, Ψ as in (34). Put $R = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $L = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$, $F = \operatorname{diag}(1, -1)$, $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $W = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = SF$. The form $Q_M = [c,\, d-a,\, -b]$ has content g, and $Q_M/g = [A', B', C']$ has discriminant $D = (t^2 - 4)/g^2$; O_D is the order of discriminant D. For $t > 2$, $w(M)$ denotes the cyclic word in R, L of M (input (K3) below); σ swaps the two letters, ρ reverses the word, and $w \sim w'$ means equality up to rotation. A class is

primitive if M is not a proper power, i.e. its word is aperiodic. Following Sarnak [25], M is *inert* if $M \sim FMF$, *reciprocal* if $M \sim M^{-1}$, and *ambiguous* if $M \sim FM^{-1}F$, all conjugacies in $SL_2(\mathbb{Z})$. A zero $\Psi(M) = 0$ is called *accidental* if M is neither inert nor reciprocal. Classes with $t < -2$ are handled through $-M$: $\Psi(-M) = \Psi(M)$, and the three symmetries commute with negation.

The following inputs are cited, not re-proved.

(K1) Ψ is an $SL_2(\mathbb{Z})$-class function on hyperbolic elements [24, Ch. 4].

(K2) $\Psi(M^{-1}) = -\Psi(M)$ [13, Lem. 6.1(ii)].

(K3) Every M with $t > 2$ is $SL_2(\mathbb{Z})$-conjugate to a positive word in R and L containing both letters, unique up to rotation; M is primitive iff the word is aperiodic. This is the cyclically reduced normal form in $PSL_2(\mathbb{Z}) \cong C_2 * C_3$; see [16, pp. 268,272] for the free-product and left-right coding.

(K4) $\Psi(M) = \#R - \#L$, the alternating sum of the run lengths of $w(M)$ [13, Prop. 4.1]; see also Zagier [27].

(K5) *(Context only.)* $-\Psi(M)/3$ is the Atiyah–Patodi–Singer eta invariant of the mapping torus T_M and its adiabatic limit, in the normalization of [6, eq. (5.60)] as restated in [13, Sec. 4.2]; $\Psi(M)$ is also the linking number of the modular knot of M with the trefoil [16]. The statement "T_M has vanishing eta invariant" below refers to this cited normalization only, not to arbitrary metrics or other operators.

(K6) For Anosov M, T_M admits an orientation-reversing self-homeomorphism iff there is an integral X with $\det X = -1$ and $XM = MX$, or with $\det X = +1$ and $XM = M^{-1}X$ [26, Sec. 1.3 and the Anosov case of the degree theorem].

(K7) Sarnak's counts by trace [25, Thm. 2] and the Sarnak–Mozzochi winding law as stated in [13, Cor. 1.3]; used only in Theorem 8.2(vi)(b).

Two elementary facts: $\Psi(FMF) = -\Psi(M)$ (eq. (35)); and $WRW = L$, $WLW = R$, $SR^{-1}S^{-1} = L$, $SL^{-1}S^{-1} = R$.

Lemma 8.1 (Witness dictionary)**.** *Let $t > 2$ and $w = w(M)$.*

(a) The following are equivalent: (1) there is $X \in M_2(\mathbb{Z})$ with $\det X = -1$ and $XM = MX$; (2) $P^{-1}MP = FMF$ for some $P \in SL_2(\mathbb{Z})$; (3) $\sigma(w) \sim w$; (4) $x^2 - Dy^2 = -4$ has an integer solution, i.e. O_D has a unit of norm -1.

(b) M is reciprocal iff $\sigma\rho(w) \sim w$.

(c) M is ambiguous iff $\rho(w) \sim w$.

Proof. (1)⇔(2): given X put $P = XF$; given P put $X = PF$; in both directions $X^{-1}MX = M \iff (XF)^{-1}M(XF) = FMF$. (2)⇔(3): $WMW = S(FMF)S^{-1}$, and WMW has the word $\sigma(w)$; apply (K3). (1)⇔(4): write $M = \begin{pmatrix} (t - gB')/2 & -gC' \\ gA' & (t + gB')/2 \end{pmatrix}$. The rational centralizer of M is $\mathbb{Q}[M]$, and its integral elements are $X(x, y) = \begin{pmatrix} (x - B'y)/2 & -C'y \\ A'y & (x + B'y)/2 \end{pmatrix}$ with $x, y \in \mathbb{Z}$, $x \equiv B'y \pmod 2$ (y is integral because $[A', B', C']$ is primitive), and $\det X(x, y) = (x^2 - Dy^2)/4$. (b) Inversion reverses the letters, and conjugation by S then gives the word $\sigma\rho(w)$ for M^{-1}. (c) $FM^{-1}F \sim WM^{-1}W$, and conjugating by S gives the word $\rho(w)$. □

8.2 What a zero of Ψ detects and forgets

Theorem 8.2. *(i)* (Witnessed zeros.) *If M is inert or reciprocal then $\Psi(M) = 0$. Ambiguity alone forces nothing: R^2L is ambiguous and $\Psi(R^2L) = 1$.*

(ii) (The inert witness is not necessary.) *$M_1 = RLR^2L^2 = \begin{pmatrix} 12 & 5 \\ 7 & 3 \end{pmatrix}$ has $\Psi(M_1) = 0$ and $D = 221$; the order O_{221} has no unit of norm -1, so no integral matrix of determinant -1 commutes with M_1. Its zero is witnessed by reciprocity: $P = RLS = \begin{pmatrix} 1 & -2 \\ 1 & -1 \end{pmatrix}$ satisfies $P^{-1}M_1P = M_1^{-1}$.*

(iii) (Neither inert nor reciprocal witness is necessary.) *$M_2 = L^3R^2LR^2 = \begin{pmatrix} 3 & 8 \\ 10 & 27 \end{pmatrix}$ has $\Psi(M_2) = 0$ and $D = 224$; it is neither inert nor reciprocal. (It is ambiguous: reversing $LLLRRLRR$ gives a rotation of it. Ambiguity does not negate Ψ and does not force the zero.) By (K6) the Sol manifold T_{M_2} is chiral, that is, it admits no orientation-reversing self-homeomorphism, while by (K5) its eta invariant, in the cited normalization, vanishes.*

(iv) (Minimality.) *Every primitive balanced class of word length ≤ 6 is inert or reciprocal. Word length 8 has exactly 8 primitive balanced classes: 2 inert, 4 reciprocal and not inert, and 2 accidental, the σ-pair $\{L^3R^2LR^2,\ L^2RL^2R^3\}$.*

(v) (Delimited converse.)

- *(a) For two-block classes R^aL^b $(a, b \geq 1)$: $\Psi = a - b$, and $\Psi = 0 \iff a = b \iff$ inert. Here the converse holds exactly.*
- *(b) For primitive four-block classes $R^aL^bR^cL^d$ $(a, b, c, d \geq 1,\ (a, b) \neq (c, d))$: $\Psi = 0 \iff a + c = b + d$; no such class is inert; the class is reciprocal iff ($a = d$ and $b = c$) or ($a = b$ and $c = d$); every other solution of $a + c = b + d$ is an accidental zero. In particular $R^2LR^{n-1}L^n$ is an accidental zero for every $n \geq 3$, with $n = 3$ the class of M_2. For unrestricted four-block tuples (a, b, c, d), the class is inert iff $a = b = c = d$.*

(vi) (Density.)

- *(a) Let B_{2n} be the number of primitive balanced classes of word length $2n$ and W_{2n} the number of those that are inert or reciprocal. Then*

$$B_{2n} = \frac{1}{2n} \sum_{d|n} \mu(d) \binom{2n/d}{n/d} \sim \frac{4^n}{2n\sqrt{\pi n}}, \qquad W_{2n} < 2^{n+1},$$

  so $W_{2n}/B_{2n} \to 0$.

- *(b)* (Cited, not computed.) *By (K7), inert or reciprocal primitive classes of trace $\leq x$ number $O(x)$, while primitive classes with $\Psi = 0$ number $\sim x^2/(12(\log x)^2)$; so in the ordering by trace as well, almost every zero is accidental.*

(vii) (What is detected.) *$\Psi(M) = 0$ detects exactly the balance $\#R = \#L$ of the cutting sequence (K4), equivalently zero winding about the cusp, zero linking with the trefoil, and vanishing eta invariant of T_M in the cited normalization (K5). The two Ψ-negating symmetries, reflection of the fibre (inert) and time reversal of the geodesic (reciprocal), force it. It does not reveal whether either of them is present.*

(viii) (Transfer of the silver zero.) *For every $J \geq 2$, every divisor $p > 1$ of $2J^2 - 1$ and $m \in \{2p, 4p\}$, the matrix M of Lemma 7.8 satisfies $-M \sim R^2L^2$, the inert two-block class of (v)(a), whose determinant -1 commuter is $\begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}$, i.e. $1 + \sqrt{2}$. The lens space $L(p, J)$ is chiral and, for $p \neq 7$, unbalanced (Proposition 7.12); so for $p \neq 7$ its zero has neither of the two named lens-space mechanisms (orientation reversal, level balance), yet it is the image of a zero that is witnessed on the Sol side. At $p = 7$ the zero is also level-wise balanced.*

Proof. (i) is (K1), (K2) and (35); $\Psi(R^2L) = 2 - 1 = 1$ by (K4), and R^2L is ambiguous because $\rho(RRL) = LRR \sim RRL$.

(ii) $\Psi(M_1) = 3 - 3 = 0$ by (K4); directly, $Q_{M_1} = [7, -9, -5]$, $s(3, 7) = -1/14$, $\Phi = 15/7 + 6/7 = 3$ and $\Psi = 0$. A unit $\eta > 1$ of O_{221} is $(x + y\sqrt{221})/2$ with $x, y \geq 1$. If $y \geq 2$ then $\eta > 15 > \epsilon := (15 + \sqrt{221})/2$; if $y = 1$ then $x^2 = 221 \pm 4$ forces $x = 15$ and norm $+1$. So ϵ is the least unit > 1, every unit is $\pm\epsilon^k$, and all have norm $+1$; by Lemma 8.1(a) M_1 is not inert. Direct multiplication gives $P^{-1}M_1P = \begin{pmatrix} 3 & -5 \\ -7 & 12 \end{pmatrix} = M_1^{-1}$.

(iii) $\Psi(M_2) = 4 - 4 = 0$; $Q_{M_2} = [10, 24, -8]$, content 2, $D = 224$. Not inert: $x^2 - 224y^2 = -4$ would give $x^2 \equiv 3 \pmod 7$, a non-residue. Not reciprocal: $\sigma\rho(LLLRRLRR) = LLRLLRRR$ has cyclic R-runs of lengths $\{1, 3\}$, whereas $LLLRRLRR$ has R-runs $\{2, 2\}$, so the former is not a rotation of the latter. Ambiguous: $\rho(LLLRRLRR) = RRLRRLLL$ is the rotation by three places. The last two sentences are (K6) and (K5).

(iv) A primitive balanced word of length ≤ 6 is a rotation of RL, R^2L^2, $RLRL$ (not primitive), R^3L^3, R^2LRL^2, R^2L^2RL, or RLR^2L^2 and their letter swaps; each is inert or reciprocal by Lemma 8.1. At length 8, classifying by cyclic run count gives one two-run class (R^4L^4, inert), four primitive four-run classes (two reciprocal, two accidental) and three six-run classes (one inert, two reciprocal); the accidental pair is $\{L^3R^2LR^2, L^2RL^2R^3\}$. Proper powers inherit the witness of their primitive root, so the first accidental length is 8 even among all balanced words.

(v)(a) $R^aL^b = \begin{pmatrix} 1 + ab & a \\ b & 1 \end{pmatrix}$, $\Psi = a - b$ by (K4), and $\sigma(R^aL^b) = R^bL^a \sim R^aL^b$ iff $a = b$. (b) Primitivity of a four-block word is exactly $(a, b) \neq (c, d)$. The R-first exponent sequences of one necklace are the rotations by an even number of places, i.e. (a, b, c, d) and (c, d, a, b). Now $\sigma(w) \sim R^bL^cR^dL^a$ has sequence (b, c, d, a), which equals one of the two only if $a = b = c = d$, excluded by primitivity (and giving the unrestricted statement); $\sigma\rho(w) = R^dL^cR^bL^a$ has sequence (d, c, b, a), which equals (a, b, c, d) iff $a = d$, $b = c$, and equals (c, d, a, b) iff $a = b$, $c = d$; both patterns satisfy $a + c = b + d$. For $(2, 1, n - 1, n)$: $a + c = b + d = n + 1$, $a = d$ would force $n = 2$, $a \neq b$, and $(2, 1) \neq (n - 1, n)$.

(vi)(a) The count of aperiodic necklaces with n letters of each kind is the displayed Möbius sum. If $\sigma(w) = \mathrm{rot}_k(w)$ with $k \not\equiv 0$, then $w_{i+k} = 1 - w_i$, so w is determined by its values on the $\gcd(2n, k) \leq n$ cycles of $i \mapsto i + k$; if $\sigma\rho(w) = \mathrm{rot}_k(w)$ then $w_{k-i} = 1 - w_i$, and the reflection $i \mapsto k - i$ has no fixed points, so it pairs the $2n$ positions. Either way at most 2^n words per k, fewer than $4n \cdot 2^n$ words in all, and each primitive class contains $2n$ words. (b) is cited.

(vii) restates (K4), (K5), (i) and (iii).

(viii) With $h = \hat{\lambda} \equiv 1 \pmod 4$, $h^2 + 6h + 1 \equiv 8 \pmod{16}$, and p divides the same number, so $8p$ divides it and both b and m are even while $-6 - h$ and h are odd: $-M \equiv I \pmod 2$, and $\mathrm{tr}(-M) = 6$. A positive word with four or more cyclic runs has trace $\geq \mathrm{tr}((RL)^2) = 7$, so a trace-6 word is R^aL^b with $2 + ab = 6$: one of R^4L, R^2L^2, RL^4. Of these only $R^2L^2 = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}$ is

$\equiv I \pmod 2$, a conjugacy-invariant condition. Hence $-M \sim R^2L^2$. (Directly, $X = (M + 5I)/2$ is integral, commutes with M and has determinant $(1 - 30 + 25)/4 = -1$.) Chirality of $L(p, J)$: $J^2 \equiv -1$ with $2J^2 \equiv 1$ would force $p \mid 3$. □

Remark 8.3. The class R^2L^2 contains, for every admissible (J, p) and both $m \in \{2p, 4p\}$, the matrix $-M$ of Lemma 7.8; this is the precise sense in which the zero of Theorem 7.1 is a "witnessed" zero on the Sol side while, for $p \neq 7$, having no orientation-reversing or level-balance mechanism on the lens-space side (at $p = 7$ it is level-wise balanced, Proposition 7.12). Neither statement transfers symmetries between the two geometries; the zero itself cannot identify why it is zero.

9 The silver-rank zero set

At a silver rank η_J is an even integer at every modulus (Corollary 6.4(b) and Theorem 6.8(c)), and its zeros include the balanced family (b) of Theorem 4.1, the orbit $q \equiv \pm J, \pm 2J$ of Theorem 7.1, and the mirror orbit when -1 is a square modulo p. At the level $p = 2J^2 - 1$ itself the zero set is the subject of the companion paper [2]. We record its main statements without proofs; the numbering refers to [2], and nothing else in this paper depends on them.

- *The S-unit coordinate* [2, Lem. 4.1, Prop. 4.3]. Reading $\mathbb{Z}[\sqrt{2}]$ modulo p through $\sqrt{2} \mapsto 2J$, the four zero families of [1, Thm. 7.1] are the orbits of $\sqrt{2}$, $\delta/\sqrt{2}$, δ and 2δ, $\delta = 1 + \sqrt{2}$. In the Cayley coordinate $\rho = (1 + q)/(1 - q)$ the family (b), the orbit of Theorem 7.1 and the mirror orbit are $\rho \in \{\pm\delta^{\pm 1}\}$, $\rho \in \{\pm\delta^{\pm 2}\}$ and $\rho^2 \equiv -1$.
- *A fifth family* [2, Thm. 5.1]: $\eta_J(L(2J^2 - 1,\, 3 - 4J)) = (J - 1)\mathbf{1}_{J \text{ odd}}$.
- *The Pell ladder* [2, Thm. 6.1]. With $\delta^n = a_n + P_n\sqrt{2}$ and $q_n = a_n + 2P_nJ$: if $P_n \mid J$ then $\eta_J(L(p, q_n)) = 0$ for odd n and $\frac{2}{3}P_{n-2}J$ for even n; if $n \geq 2$ and $J \equiv 1 \pmod{P_n}$ then $\eta_J(L(p, q_n)) = \gamma_n(J - 1)/P_n$ for an explicit $\gamma_n > 0$.
- *The period law* [2, Conj. 7.1, Thm. 8.4, Cor. 8.5]. Conjecturally every nonconstant S-unit orbit is eventually quasi-polynomial in J with period dividing its coefficient of $\sqrt{2}$, supported by a reported out-of-sample test on 64 orbits (only summary counts were retained; see the companion's Appendix A). This is proved for every nonconstant orbit of norm ± 1 or ± 2 (the powers δ^n, $n \geq 1$, and the norm-two orbits $a_nJ \pm P_n$), with explicit thresholds past which η_J is linear in J on each residue class, complete zero sets for 24 families, and eventually periodic zero sets; the constant orbit $q = \pm 1$ lies outside the law, and there $\eta_J(L(2J^2 - 1, 1)) = -J(J - 1)(2J - 1)/3$ is cubic.
- *Rank-uniform zeros* [2, Cor. 8.9, Cor. 7.2(a)]. Among the orbits of norm ± 1 and ± 2 exactly $q \equiv J,\ J + 1,\ 2J - 1$ vanish at every rank; in a window of 64 orbits exactly the four families of [1] do.
- *Unbounded ambiguity* [2, Cor. 8.13]. For every K there is a rank J at which K pairwise non-homeomorphic chiral lens spaces $L(2J^2 - 1, q_i)$, none rank-J balanced (Theorem 4.1), all have $\eta_J = 0$.

The last item is the sharp form of the gap between vanishing and balance of Example 4.5. By Corollary 4.4 a zero of η_1 is exactly a mirror symmetry, and by Corollary 6.4(c) the same holds at prime moduli off the silver ranks with $p \nmid J$; at silver ranks $J \geq 2$ a single vanishing η_J can hide an unbounded amount of handedness. The companion also records a refuted conjecture [2,

Conj. 4.6] and the reason the Rademacher mechanism of Section 7 does not close the classification [2, Rem. 4.5].

10 Open problems

(1) *Silver-rank zeros.* Determine, for $p \mid 2J^2 - 1$, the set of q with $\eta_J(L(p,q)) = 0$ (Section 9 and [2, Sec. 9]); in particular prove the period law [2, Conj. 7.1] beyond norm ± 2, and describe the zeros outside the S-unit sector.

(2) *Composite moduli.* Theorem 6.8(b),(c) gives necessary local conditions for a zero; they are not sufficient ($\eta_{25}(L(25,1)) = -104$). Observation 5.7 shows zeros in which several nonzero primitive parts cancel. A classification of the zeros of η_J at composite p, even off the silver ranks of the prime factors, is open.

(3) *The prime 3 and $p = 2, 3$ boundaries.* Theorem 6.3 fails at $p = 3$ ($\eta_1(L(3,1)) = -2/9$) and Theorem 6.8(d) needs $3 \nmid p$; the 3-primary behaviour of η_J is described only by Theorem 6.8(c).

(4) *Literature.* Congruences of the type of Theorems 6.3 and 6.7 may be instances of index-theoretic congruences for flat bundles; and Corollary 7.11 is likely classical. Neither has been searched for.

(5) *Other operators.* Whether the level-wise criterion of Theorem 3.3 has an analogue for twisted or half-integer spin operators on lens spaces, where the eta invariants are known [7, 11], is not addressed here.

A Exact computations

Every statement in this paper was accompanied by exact computations, using integer or rational arithmetic (or certified interval arithmetic where a transcendental identity is involved), and in several cases by predictions fixed before the corresponding rows were computed ("held-out" below; "opened" rows were available when the statement was formulated). These computations test the identities and the conventions; they are not substitutes for the proofs. Numerical values quoted as "worst difference" are high-precision floating-point corroboration, not interval enclosures.

statement	what was checked	rows	failures
Branch counts (3)	direct weight-pair enumeration against a Clebsch–Gordan recurrence on residue-class character vectors, $p \le 40$, all q, $J \in \{1,2\}$, levels $\le 2p+10$	63,198 (opened)	0
Eta anchor	the regularized level sum equals η_J from the master formula (6) as emitted by an independent certified implementation, $p \le 30$, all q, $J \in \{1,2,3\}$	831 (opened)	0
Corollary 3.2	periodic increments of residue-class weight counts, $p \le 60$, every multiplier; windows $n < 2p$ and $n < 6p$ give identical verdicts, $p \le 40$, $J \le 4$	all rows	0
Corollary 4.4	rank-one balance $\iff q^2 \equiv -1$, $p \le 60$; $s(h,k) = 0 \iff h^2 \equiv -1$, $k \le 300$	all rows	0

statement	what was checked	rows	failures
Rank 4 prediction	chiral balanced spaces at $J = 4$, $p \le 64$, predicted as $p \in \{31, 62\}$ with four q each before computation	held-out	hit
Theorem 3.1	(10) evaluated in 200-bit interval arithmetic contains the exact integer $pD_J(n)$, $p \le 30$, all q, $J \le 8$, $n \le 2p$	91,840 (opened)	0
Theorem 3.3	criterion (i)$\wedge$(ii) equals direct balance on $n < 6p$, $p \le 260$, all q, $J \le 8$; the symbolic identities of the proof verified in a computer algebra system	164,784	0
Example 4.6	chiral balanced sets for $J \le 8$, $p \le 260$, and eight rank scans, predicted from the families before the rows $J \ge 5$ were computed	held-out	hit
Theorem 4.1 (support)	criterion $\iff$ (a)$\vee$(b)$\vee$(c) for all $p \le 300$, all q, all J mod p	5,483,516	0
Lemma 3.9	balance of $L(2p, \tilde{q})$ equals balance of $L(p, q)$ by the criterion, odd $p \le 151$, all q, $J \le 2p$	954,920	0
Remark 4.2	class-generator membership over $\mathbb{Z}$ in the unit-saturated ideal, tested for each of the 105 matchings (one representative basis per class retained); semantic check on every odd $p \in [5, 101]$ and the composites 121, 125, 143, 169, 175, 187, 221, 245, 289, 323, 343, 385 (all units q, all J mod p); even reduction for $p \equiv 2 \pmod 4$, $p \le 402$	105; 712,060; 1,116,462	0
Theorem 5.2	primitive parts by the element route (17)–(18) against Möbius inversion, $d \le 60$, all q, $J \le 8$ (worst difference $1.7 \cdot 10^{-55}$ at 60 digits)	8,808	0
Proposition 5.4	$\sum_r u_r = 0$ and $\sum_r [v_r - a_r u_r] = 0$, $p < 40$, all q, $J \in \{1, 2, 3, 5\}$	1,888	0
Observation 5.5	zeros of η_J on primes 41–199 ($J \le 12$) and 41–101 ($13 \le J \le (p-1)/2$); the prediction, fixed in advance, "zeros are mirror, silver-family, or the four families at $p = 2J'^2 - 1$" was refuted by 196 zeros in 49 orbits	71,230 (held-out)	196 unpredicted
Observation 5.7	zeros on composites 41–120, $J \le 10$, classified by primitive parts	25,180	–
Observation 5.6	dichotomy, silver integrality and $q = J$ on primes 211–599 never inspected before the statements were fixed	67,032 + 41,286 + 12,210 (held-out)	0
Theorem 6.3	symbolic Laurent coefficients of f, $f_{J,1}$ and the trace constants (two independent computer algebra systems); trace form (20) against $p\,\eta_J$, primes 5–47, all q, $J \le 6$ (worst difference $8.7 \cdot 10^{-57}$); (i), (ii), Corollary 6.4(a) on all 17 primes 601–701, all q, $J \in \{1, 2, 3\}$, plus silver ranks $J_s, J_s + 1$ and $J = p$ at 601, 607; master formula against the evaluator, $p \le 40$, $J \le 8$; Rademacher's congruence, primes 5–199	1,860; 43,026 (held-out); 3,912; 4,178	0

statement	what was checked	rows	failures
Corollary 6.4(d)	$\eta_{np}(L(p,\pm1)) = \mp n(p^2-1)/6$ at $p \in \{5,7,11,13\}$, $n \in \{1,2\}$; $\eta_J - J(2J^2-1)\eta_1 \in 2\mathbb{Z}$ on primes 5–59, all q, $J \le 12$	8; 5,040	0
Theorem 6.7	master formula against the evaluator, $41 \le p \le 60$, $J \le 8$; Lemma 6.2, $p \le 400$; Lemma 6.1, $p \le 80$, $J \le 16$; (30) on every composite 121–220 ($J \le 6$), on $p \in \{2,3\}$ ($9 \le J \le 60$) and on $p \in \{4,8,9,16,25,27,32,49\}$ ($9 \le J \le 24$); Theorem 6.8(b) on the 364 zeros found	4,896; 48,677; 31,440; $45{,}072 + 156 + 1{,}856$ (held-out except $p = 121, 150, 200$ and $p \in \{2,3\}$, which were computed earlier)	0
Theorem 7.1	the chain $K = 0$, $16K = T - p$, parity split, Lemma 7.7 at $m \in \{2p, 4p\}$, Lemma 7.8 by two Dedekind-sum routes, $\Psi(M) = 0$ and content 2, on every divisor $p > 1500$ of $2J^2 - 1$ for $60 \le J \le 110$ and at both silver ranks of the first 12 primes $\equiv 1$ and 12 primes $\equiv 7 \pmod 8$ above 100,000 (held-out), and on $p \le 1500$, $60 \le J \le 110$ (opened); $\eta_J(L(p,J)) = 2K/p^2 = 0$ by the exact evaluator for $p \le 12000$; Corollary 7.11 on every row; family (b) on 15 rows	$62 + 48$ (held-out) +73; 29; 183; 15	0
Theorem 8.2	(K4), (35), (K2), word route against the Gauss reduced-form route, inert $\iff$ negative Pell, on every primitive word of length ≤ 18; M_1, M_2; all primitive balanced classes of length ≤ 8; two-block $a, b \le 60$; primitive four-block tuples with $a+b+c+d \le 32$ (1,232 zeros, 1,008 accidental); the family $(2, 1, n-1, n)$, $n \le 60$; B_{2n} and W_{2n} at every even length ≤ 24 ($B_{22} = 32{,}065$, $W_{22} = 1{,}085$; $B_{24} = 112{,}632$, $W_{24} = 2{,}150$); $-M \sim R^2L^2$ for $J \le 40$, all divisors, both m	31,040; 13; 3,600; 35,840; 58; 12 lengths; 154	0
Section 9	the exact computations behind the companion's statements (fifth family, Pell ladder, period-law test and proof, complete zero sets, the refuted Conjecture 4.6) are listed in [2, App. A]	–	–

Share of primitive balanced classes with an inert or reciprocal witness, W_{2n}/B_{2n}: 6/8 at length 8, 29/75 at 12, 128/800 at 16, 531/9,225 at 20, 1,085/32,065 at 22, 2,150/112,632 at 24.

In each block, deliberately mutated versions of the statements (a wrong sign, a dropped term, a wrong root, a wrong residue class) were run and failed at a named witness; these mutation checks guard against a checker that passes vacuously. One refutation of a prediction fixed in advance is recorded above: the conjectured prime-modulus zero list was false, and the zeros it missed led to Observation 5.5 and thence to Theorems 6.3 and 7.1. Two further refutations, of the companion's Conjecture 4.6 at $p = 1129$ and of a control premise at $q = 4 - 4J$, are recorded in [2, App. A].

Authorship, assistance and rights

Luqman Mahmood directed this work and is responsible for its claims and final text. OpenAI Codex and Anthropic Claude assisted with mathematical development, computation, checking and editing. Internal and automated checks are not external peer review. Computational tests do not replace proofs of universal statements, and no physical identification or experiment is implied. The paper and mathematical data are licensed CC BY 4.0. Software source and executable mathematical verification scripts are excluded from that grant; their rights are reserved unless explicitly stated otherwise.

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